

Bayesian Ranking and Graph-Based Review Design under Sparse Evaluation Data

A Case Study of the CRS Seed Grants 2026

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Abstract

Grant peer review often requires funding decisions based on only a small number of evaluations per proposal, making the uncertainty of resulting rankings difficult to assess. This report studies the 2026 CRS Seed Grant evaluation, where 42 proposals were each reviewed twice by a panel of nine reviewers. A Bayesian hierarchical ranking model separates proposal effects, reviewer-associated scoring tendencies and residual review-level variation, while representing ranking uncertainty through posterior rank distributions.

A CRS-specific cross-classified model reproduced the four proposals selected under the final CRS decision rule, but substantial uncertainty remained in the ordering of individual proposals. The model was computationally stable and insensitive to the tested prior specifications. Posterior predictive checks showed good agreement with observed review patterns, including average reviewer disagreement, while the unbounded Gaussian representation of the five-point scores remained an explicit modelling caveat.

Finally, the reviewer–proposal assignment structure was used to study where additional reviews could be most informative. Although different statistical criteria preferred somewhat different individual assignments, a practical comparison of small additional-review budgets found very little difference in global uncertainty reduction between random allocation, targeting proposals with high reviewer disagreement and model-based sequential allocation.

Overall, the results show how Bayesian ranking can complement transparent evaluation rules by making ranking uncertainty explicit. They also suggest that the value of targeted additional reviews depends on the purpose of the extra evaluation: in the present CRS design, carefully optimizing a small number of assignments offered little advantage for reducing global ranking uncertainty.

1 Introduction

1.1 Grant evaluation under limited information

Grant peer review converts a limited number of expert judgments into decisions that can have substantial consequences for applicants. The challenge is not only disagreement between reviewers, but also the risk of overinterpreting small score differences when each proposal is assessed by only a few reviewers. Kaplan

et al. show directly that finer scoring precision requires increasingly many independent reviewers, and illustrate how rankings can vary strongly when only a small number of evaluations are available ([Kaplan et al. 2008](#)).

This uncertainty becomes particularly important when proposals are close in score or close to a decision boundary. In a retrospective analysis of Australian grant panels, Graves et al. found substantial instability in funding status after accounting for variability among panel members’ scores ([Graves et al. 2011](#)). Jayasinghe, Marsh and Bond reported low single-rater reliability in Australian Research Council grant assessments and used a multilevel cross-classified model to separate proposal- and assessor-related sources of variation ([Jayasinghe et al. 2003](#)). Heyard et al. similarly motivate Bayesian ranking by low inter-rater reliability and by the difficulty of distinguishing proposals of similar quality near a funding line ([Heyard et al. 2022](#)). These findings do not imply that peer review is uninformative; rather, they motivate distinguishing an observed numerical ordering from the strength of evidence supporting that ordering.

The present project examines this distinction in a small grant-evaluation setting and subsequently considers whether the structure of the review design itself can be used to improve the allocation of limited review effort.

1.2 CRS Seed Grant evaluation 2026

The Center for Reproducible Science and Research Synthesis (CRS) at the University of Zurich (UZH) works to advance scientific methods and practices that support trustworthy and reproducible research. Its activities include research on reproducibility, research synthesis, and meta-research ([Center for Reproducible Science and Research Synthesis 2026b](#)).

In 2026, the CRS launched its first Seed Grant call to support early-stage projects and initiatives in research synthesis, meta-research, and related fields ([Center for Reproducible Science and Research Synthesis 2026b](#)). The evaluation round included 42 proposals assessed by a panel of nine reviewers. Each proposal received exactly two independent reviews, resulting in 84 completed review records.

Reviewers evaluated each proposal on four supporting criteria: scientific quality; potential for scaling and obtaining substantial third-party funding; relevance to fostering a sustainable, innovative, and diverse Swiss “research on research” community; and commitment to good research practices, open science, and reproducibility. The original call described selection using the average score across these criteria and reviewers ([Center for Reproducible Science and Research Synthesis 2026b](#)). During the development of the evaluation procedure, this was subsequently replaced by a separate holistic overall grade on a five-point scale, which became the score used for qualification. The reason for this change was not documented in the material available for this project.

The call indicated that approximately six projects could be funded ([Center for](#)

[Reproducible Science and Research Synthesis 2026b](#)). In the final evaluation procedure, four proposals met the qualification rule and were selected for funding, as documented on the published results page ([Center for Reproducible Science and Research Synthesis 2026a](#)). The rule is reconstructed explicitly in Section 2.3.

1.3 Motivation for Bayesian ranking

The final CRS qualification procedure was transparent and easy to apply, but it did not quantify how strongly the available reviews separated the underlying qualities of different proposals. Observed grades may also reflect both proposal quality and systematic differences in reviewer scoring behaviour.

Bayesian hierarchical modelling provides a way to represent these sources of variation separately. Heyard et al. model latent proposal effects together with assessor-related effects and rank proposals using the posterior distribution of the proposal effects rather than the raw score averages alone ([Heyard et al. 2022](#)). Their expected-rank approach retains uncertainty about the relative ordering instead of treating an estimated ranking as known with certainty.

For the CRS data, Bayesian ranking is therefore used to quantify uncertainty and support interpretation of the evaluation rather than as a replacement rule that mechanically determines funding.

1.4 Research questions

1.4.1 Research Question 1

What ranking does the Bayesian model produce after accounting for reviewer differences, and how uncertain are the resulting ranks?

1.4.2 Research Question 2

How does the main Bayesian ranking model compare with the final CRS qualification rule and the ranking based on the observed overall grades?

1.4.3 Research Question 3

Is the main Bayesian model computationally stable, adequate under posterior predictive checks, and robust to the tested prior specification?

1.4.4 Research Question 4

How closely do the four supporting criterion scores correspond to the holistic overall grade, and would an equal-weight average of the criteria have produced a materially different proposal ordering?

1.4.5 Research Question 5

Which single additional reviewer–proposal assignment would most reduce uncertainty under the fitted Bayesian model, and how does this depend on the chosen notion of uncertainty?

1.5 Scope and structure of the report

The holistic overall grade is the response for the main Bayesian analysis. The four supporting criteria are examined separately, including an equal-weight criterion average used as a descriptive sensitivity analysis rather than as a preferred scoring rule. V0 reproduces the continuous Heyard reference specification, while the simpler cross-classified V1 model is used as the main CRS model. After comparing the resulting rankings, Chapter 5 evaluates the computational stability, posterior predictive behaviour and prior sensitivity of V1.

Chapter 6 then treats the reviewer–proposal assignment structure as a statistical-design problem. It first asks which one currently unobserved assignment would most reduce posterior uncertainty under V1 and compares A-, D- and E-optimality criteria. It then extends the primary A-optimality criterion to a practical comparison of small additional-review budgets under random, disagreement-targeted and sequential A-optimal allocation. Because reviewer expertise, conflicts and eligibility are not available, these analyses are interpreted as statistical design comparisons rather than operational assignment recommendations. The final discussion brings together the ranking, validation, criterion and review-design results and derives practical implications for future CRS evaluations.

2 Evaluation design and descriptive analysis

2.1 Evaluation process and analytical data

The pseudonymized analytical dataset contains 84 completed reviews of 42 proposals by nine reviewers. Each proposal received exactly two independent reviews. Reviewers completed between 6 and 12 reviews, with assignments based on expertise and availability.

Each review contains four supporting criterion scores and a separate holistic overall grade. All scores use an integer scale from 1 to 5, where higher values indicate a more favourable evaluation. The Bayesian models use the holistic overall grade as their outcome. The four criterion scores are retained for a separate descriptive analysis of their relationship with the overall grade.

2.2 Reconstruction of the final qualification procedure

According to the published CRS Seed Grant results ([Center for Reproducible Science and Research Synthesis 2026a](#)), funding was available for proposals that received the highest overall grade from at least one reviewer and no grade below

the second-highest category. Since the overall grade was scored on a scale from 1 to 5, this means that a proposal qualified if it received at least one grade of 5 and its second grade was at least 4. With exactly two reviews per proposal, the qualifying score combinations were therefore (5,5), (5,4), and (4,5). Equivalently, a proposal qualified if its mean overall grade was at least 4.5.

This procedure should be interpreted as a score-based qualification threshold rather than as a fixed top-four selection rule. Four proposals satisfied the threshold in the 2026 evaluation round. Since the call indicated that approximately six projects could be funded, the available funding capacity was not binding and no additional prioritization among qualified proposals was required.

2.3 Criteria and holistic overall grade

Previous NIH analyses have examined how criterion scores relate to holistic overall evaluations (Eblen et al. 2016; Erosheva et al. 2020). Eblen et al. found substantial differences between criteria in their association with the Overall Impact score (Eblen et al. 2016). This motivates examining whether the four supporting CRS criteria correspond equally closely to the separately assigned holistic overall grade.

Let C_{rc} denote the score assigned in review r to criterion c , and let Y_r denote the holistic overall grade. Their relationship was summarized descriptively using Pearson correlation for linear association and Spearman correlation for monotonic association based on ranks. The corresponding sample definitions are given in Section 9.1. These correlations are treated descriptively because the 84 reviews are not independent with respect to proposals and reviewers.

Criterion	Mean criterion score	Pearson correlation	Spearman correlation
Scientific quality	3.643	0.892	0.890
Community building	3.179	0.545	0.551
Scaling potential	3.512	0.541	0.553
Scientific rigor	3.881	0.588	0.599

Scientific quality showed the strongest relationship with the holistic overall grade, with Pearson and Spearman correlations of 0.892 and 0.890. The remaining three criteria were positively related to the overall grade but less strongly, with correlations around 0.54–0.60. These correlations do not, however, identify the weights reviewers placed on the individual criteria.

For a simple equal-weight summary, define the criterion average for review r as

$$A_r = \frac{1}{4} \sum_{c=1}^4 C_{rc}.$$

Across the 84 reviews, the mean holistic overall grade was 3.476 and the mean equal-weight criterion average was 3.554. The Pearson correlation between A_r and Y_r was 0.859 and the Spearman correlation was 0.849. The mean difference $A_r - Y_r$ was 0.077 grade points, indicating little systematic difference in level between the two summaries, and 78.6% of reviews had a criterion average within 0.5 grade points of the holistic overall grade.

2.4 Equal-weight criterion summary at proposal level

The equal-weight summary is used as a descriptive counterfactual rather than as an estimate of how reviewers actually combined the four criteria. For proposal i , let

$$O_i = \frac{1}{2} \sum_{r:p(r)=i} Y_r$$

denote its mean holistic overall grade, and let

$$A_i = \frac{1}{2} \sum_{r:p(r)=i} A_r = \frac{1}{8} \sum_{r:p(r)=i} \sum_{c=1}^4 C_{rc}$$

denote the equal-weight average of all eight criterion scores received across its two reviews.

The two proposal-level summaries remained strongly related, with Pearson correlation 0.848 and Spearman correlation 0.881. Their relationship is shown in Figure 1.

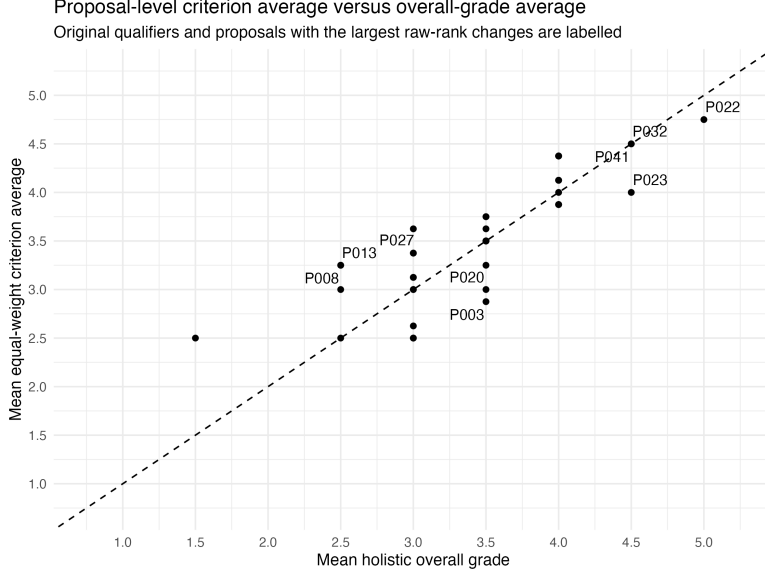


Figure 1: Proposal-level equal-weight criterion average versus mean holistic overall grade. The dashed identity line represents identical proposal-level summaries. CRS qualifiers and proposals with the largest **raw-rank changes** are labelled.

The holistic proposal average is relatively coarse because it is calculated from only two integer-valued grades, whereas A_i averages eight criterion scores and therefore has finer numerical resolution. Raw-rank changes are consequently sensitive to the breaking of ties and are not emphasized here.

At the top of the distribution, P022 had criterion average 4.75 and P032 and P041 had 4.50. These are three of the four proposals that qualified under the final CRS rule. The fourth qualifier, P023, had criterion average 4.00, while several non-qualified proposals had criterion averages of 4.375. Thus, equal weighting preserves substantial agreement with the holistic evaluation but does not reproduce its upper ordering exactly.

3 Bayesian ranking framework

3.1 Related work

The Bayesian ranking framework used in this project follows directly from Heyard et al. (2022), who developed a Bayesian hierarchical model for the evaluation of research proposals at the Swiss National Science Foundation. Their model represents observed evaluation scores through latent proposal and assessor-related effects. Rather than ranking proposals only by observed scores or posterior mean proposal effects, they use the posterior distribution of the proposal effects to

calculate expected ranks. This expected-rank formulation, which accounts for uncertainty in the relative ordering of proposals, is also the principal Bayesian ranking quantity used in this report.

Heyard et al. place this approach within an earlier literature on Bayesian ranking. In particular, they draw on Laird and Louis (1989), who developed Bayes and empirical-Bayes methods for estimating the ranks of random effects in variance-component models. Heyard et al. use their expected-rank formulation,

$$E[\text{rank}(\theta_i)],$$

as the basis for ranking the latent proposal effects.

Bayesian hierarchical modelling has also previously been applied directly to grant-review ratings. Johnson (2008) analysed NIH peer-review ratings using a Bayesian hierarchical model that allows for differences in reviewer scoring patterns and quantifies posterior uncertainty in proposal ratings. For ordinal grant scores, Heyard et al. additionally draw on Johnson (2008) and Cao, Stokes and Zhang (2010) to formulate an alternative model based on an underlying continuous proposal-quality trait that is translated into observed ordinal categories. Cao et al. develop such a Bayesian hierarchical multirater model explicitly for ranking, allowing both a latent item trait and rater-specific characteristics, and demonstrate the approach using grant-review data.

The proposal-reviewer structure is also related to the cross-classified grant-review model of Jayasinghe, Marsh and Bond (2003). When reviewers assess multiple proposals and proposals receive evaluations from multiple reviewers, proposal and reviewer effects are crossed rather than nested. This provides the structural motivation for retaining persistent proposal and reviewer effects in the simpler V1 specification used here.

More recently, Heyard et al. (2025) applied Bayesian ranking to individual evaluations of 1,006 Marie Skłodowska-Curie Actions proposals and compared the resulting model-based rankings with scores, rankings and funding decisions established after consensus meetings. Their analysis again used expected ranks of latent proposal effects while adapting the hierarchical model to the information available on individual experts.

The present analysis therefore does not introduce a new Bayesian ranking method. It takes the framework of Heyard et al. as its methodological starting point, reproduces their continuous reference specification as V0, and examines whether a simpler crossed proposal-reviewer specification preserves the substantive ranking conclusions in the CRS setting.

3.2 V0: Heyard reference model

The reference model V0 reproduces the continuous homogeneous-variance specification of Heyard et al. (2022). Let $r = 1, \dots, 84$ index completed reviews, let

$p(r)$ denote the proposal reviewed in record r , and let $a(r)$ denote the reviewer. The observed holistic overall grade is Y_r , and $\bar{Y}$ denotes its observed mean.

The model is

$$Y_r \mid \theta, \lambda \sim \mathcal{N}(\bar{Y} + \theta_{p(r)} + \lambda_{p(r),a(r)}, \sigma^2),$$

with proposal effects

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2),$$

and proposal-reviewer assessment effects

$$\lambda_{ij} \sim \mathcal{N}(\nu_j, \tau_\lambda^2), \quad \nu_j \sim \mathcal{N}(0, 0.5^2).$$

The latent proposal effect θ_i is the quantity used for ranking. The reviewer-specific mean ν_j represents a persistent scoring tendency, while $\lambda_{ij} - \nu_j$ permits an additional deviation for a particular proposal-reviewer assessment.

V0 is retained as the methodological reference model rather than as the preferred CRS specification. Each observed proposal-reviewer combination occurs only once in the CRS data, so the additional assessment-specific effect is weakly informed at the individual level. This motivates the simpler V1 formulation.

3.3 V1: CRS cross-classified model

The CRS-specific model V1 simplifies V0 by removing the proposal-reviewer-specific assessment effect and serves as the main model for the CRS analysis. Each observed grade is decomposed into three sources of variation: a proposal effect, a reviewer effect and residual review-level variation.

Let $p(r)$ denote the proposal evaluated in review r , and let $a(r)$ denote the reviewer who produced that review. V1 assumes

$$Y_r = \bar{Y} + \theta_{p(r)} + b_{a(r)} + \varepsilon_r, \quad \varepsilon_r \sim \mathcal{N}(0, \sigma^2),$$

or equivalently,

$$Y_r \sim \mathcal{N}(\bar{Y} + \theta_{p(r)} + b_{a(r)}, \sigma^2).$$

Here, $\bar{Y}$ is the observed mean overall grade. The proposal effect θ_i represents how proposal i tends to score relative to this overall mean, while the reviewer effect b_j represents how reviewer j tends to score relative to other reviewers. The remaining term ε_r captures review-level variation that is not explained by either effect.

The proposal and reviewer effects are assigned hierarchical distributions,

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2), \quad b_j \sim \mathcal{N}(0, \tau_b^2).$$

The latent proposal effects θ_i are the quantities used for ranking. Because reviewer assignments were based on expertise and availability rather than random allocation, the estimated reviewer effects are interpreted descriptively as reviewer-associated scoring tendencies rather than as causal measures of reviewer bias.

The model is cross-classified because every observed grade belongs simultaneously to a proposal and a reviewer, while neither grouping is nested within the other, meaning that reviewers are not restricted to a single proposal and proposals are not restricted to a single reviewer.

The hierarchical specification also induces partial pooling. Proposal effects with limited information are pulled toward zero, corresponding to the overall mean grade $\bar{Y}$ on the observed scale, with the amount of pooling governed by the estimated between-proposal variation τ_θ^2 . Reviewer effects are pooled analogously through τ_b^2 . This regularization is particularly relevant in the CRS setting because each proposal contributes only two observed grades.

Estimation follows from Bayesian updating: the likelihood contributed by the observed grades is combined with the hierarchical prior distributions to obtain the joint posterior distribution of the proposal and reviewer effects and the variance parameters. A short mathematical derivation of this updating and the resulting partial pooling is given in Section 9.2.

3.4 Assumptions and prior specification

Both V0 and V1 treat the observed grades as conditionally Gaussian with a common residual variance and assume exchangeable proposal effects around zero. V1 additionally treats the reviewer effects as exchangeable around zero. The overall mean $\bar{Y}$ is fixed at the observed mean grade rather than estimated as a separate intercept.

For V0, the scale parameters use

$$\sigma, \tau_\theta, \tau_\lambda \sim \text{Uniform}(0, 2),$$

while V1 uses

$$\sigma, \tau_\theta, \tau_b \sim \text{Uniform}(0, 2).$$

The lower endpoint is implemented numerically as 10^{-6} . The sensitivity of the main V1 conclusions to this scale-prior specification is examined in Chapter 5.

Although the holistic overall grade is recorded on an integer scale from 1 to 5, both models use the continuous Gaussian formulation of Heyard et al. (2022). This treats adjacent categories as equally spaced. Since the main target is the relative ranking of latent proposal effects rather than prediction of valid score categories, this approximation is retained here; its limitations are assessed through posterior predictive checks in Chapter 5.

3.5 Posterior ranking quantities

The ranking is based on the posterior distribution of the proposal effects $\theta_1, \dots, \theta_{42}$ rather than directly on the observed score averages. For posterior draw s , the rank of proposal i is

$$R_i^{(s)} = 1 + \sum_{k \neq i} \mathbb{1}_{\{\theta_k^{(s)} > \theta_i^{(s)}\}},$$

where rank 1 corresponds to the proposal with the largest latent proposal effect and $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function.

The posterior mean of this rank distribution is the expected rank,

$$ER_i = E[R_i \mid Y].$$

Using the definition of R_i ,

$$ER_i = 1 + \sum_{k \neq i} \mathbb{P}(\theta_k > \theta_i \mid Y).$$

Expected rank can therefore be interpreted as 1 plus the sum of the posterior probabilities that each competing proposal has a larger latent effect than proposal i . In practice, it is estimated from the posterior draws by

$$ER_i \approx \frac{1}{S} \sum_{s=1}^S R_i^{(s)}.$$

Expected rank is the principal Bayesian ranking quantity used in this report and follows Heyard et al. (2022). With 42 proposals, the midpoint of the rank scale is

$$\frac{42 + 1}{2} = 21.5.$$

When many of the pairwise ordering probabilities for proposal i are close to 0.5 ($\mathbb{P}(\theta_k > \theta_i \mid Y) \approx 0.5$), its expected rank is consequently pulled toward the centre of the rank scale.

Three ranking summaries are distinguished throughout the analysis. The **raw-score rank** orders proposals by their mean observed overall grade, the **posterior-mean rank** orders the posterior means of θ_i , and the **expected rank** averages the proposal’s rank across the full posterior distribution. Rank uncertainty is summarized by the 90% posterior rank interval, defined by the 5th and 95th percentiles of R_i .

For comparisons between particular proposals, pairwise posterior probabilities such as

$$\mathbb{P}(\theta_i > \theta_k \mid Y)$$

give a more direct measure of the evidence for one proposal being ranked above another. These are particularly useful when small changes in a model produce different integer ranks for proposals that are nearly tied.

3.6 Posterior computation and diagnostics

All Bayesian models were fitted in JAGS from R. V0 uses the explicit Heyard reference specification implemented through **ERforResearch**, while V1 uses a custom JAGS model with one persistent reviewer effect per reviewer.

Unless stated otherwise, each final fit used four Markov chains, 10,000 adaptation iterations, 10,000 burn-in iterations and 50,000 retained sampling iterations per chain, giving 200,000 posterior draws in total. All MCMC fits used a fixed random seed of 20260727 to ensure reproducibility.

Posterior-mean ranks, expected ranks, rank intervals and pairwise posterior probabilities were calculated from the retained posterior draws. Numerical convergence was assessed using the potential scale reduction factor (PSRF) and effective sample size (ESS). These diagnostics assess whether the posterior computation is stable; statistical model adequacy is evaluated separately through posterior predictive checks and sensitivity analysis in Chapter 5.

4 Bayesian ranking results

4.1 Main V1 ranking and comparison with the final CRS procedure

The CRS-specific V1 model preserves the broad upper ordering based on the observed overall grades while showing that the exact latent ranks are highly uncertain. P022 had posterior-mean rank 1 and expected rank 13.08. P032, P023 and P041 occupied posterior-mean ranks 2, 3 and 4, with expected ranks 15.87, 15.87 and 16.22, respectively. These are exactly the four proposals that satisfied the final CRS qualification threshold.

The posterior-mean ordering is therefore much sharper than the posterior evidence supporting the exact relative positions. The 90% posterior rank interval for P022 was 1–36; for P023 and P032 it was 1–37; and for P041 it was 1–38. The relationship between the observed-score rank, posterior-mean rank and posterior expected rank is shown in Figure 2.

This distinction is the main additional information provided by the Bayesian ranking relative to the final qualification rule. The CRS procedure identifies proposals that satisfy a transparent score threshold; it does not quantify how strongly the available reviews separate their latent proposal effects. In the present data, the same four proposals appear strongest under both approaches, but their exact positions are weakly identified. The Bayesian analysis is therefore interpreted as uncertainty-aware decision support rather than as a retrospective replacement for the CRS rule.

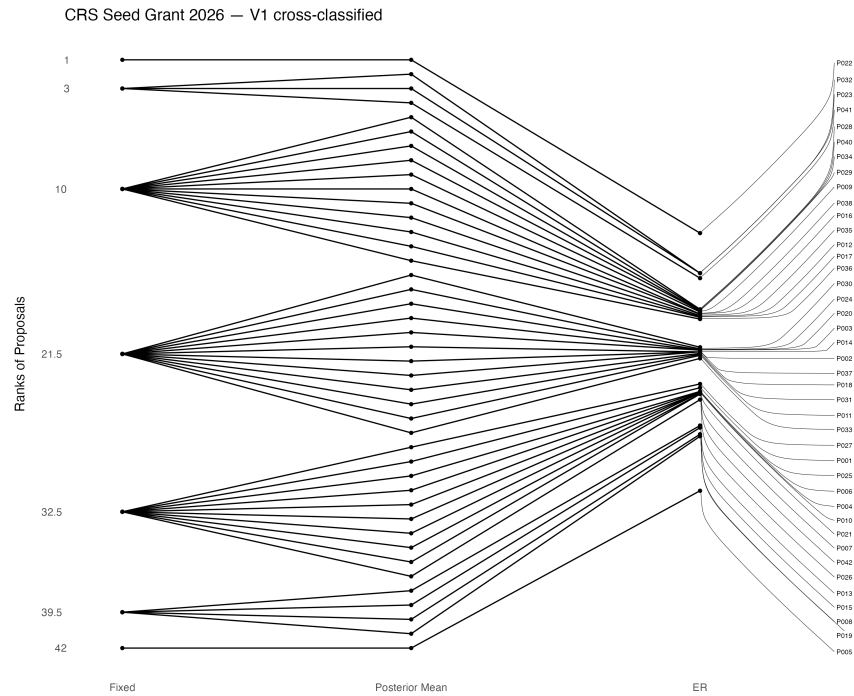


Figure 2: Proposal rankings under the observed-score ranking, posterior-mean ranking and posterior expected rank for the V1 cross-classified model. The movement of expected ranks toward the centre of the rank scale reflects substantial posterior uncertainty in the exact proposal ordering.

4.2 Robustness to the Heyard reference specification

The V0 Heyard reference model gives almost the same substantive result. The four CRS qualifiers again occupy the first four posterior-mean positions. Across all 42 proposals, the Spearman correlation between V0 and V1 posterior-mean ranks was 0.9951 and the Pearson correlation between expected ranks was 0.9916. The close agreement is visible in Figure 3.

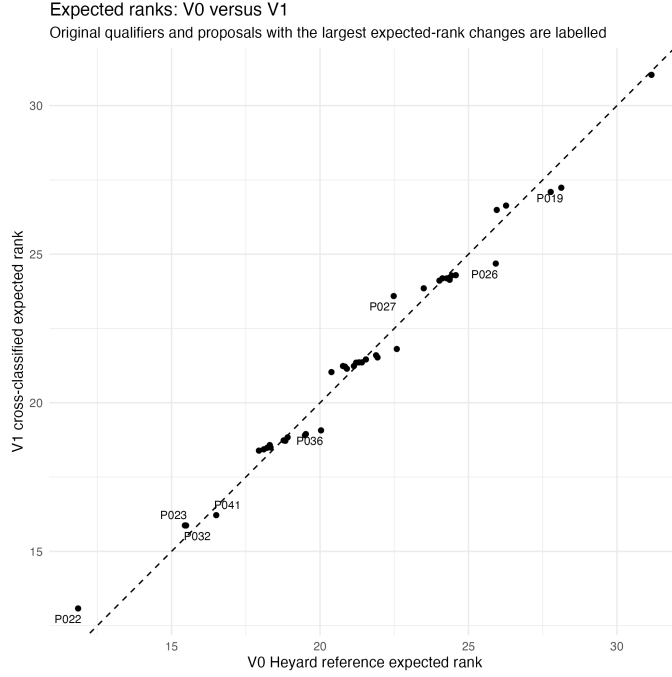


Figure 3: Posterior expected ranks under the V0 Heyard reference model and the V1 cross-classified model. The dashed identity line represents identical expected ranks under both models.

Small changes in integer ranks can nevertheless occur when proposals are almost indistinguishable. Twenty-six of the 42 proposals changed posterior-mean integer rank, but these changes were small: the mean absolute change was 0.86 positions and the maximum was three positions. P023 and P032 provide a clear example: V0 places P023 above P032, while V1 reverses them. Yet

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.498955$$

under V0 and

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.500470$$

under V1. The apparent reversal is therefore an almost exact posterior tie rather than evidence that the two models disagree meaningfully about these proposals.

Rank uncertainty is equally stable across the two specifications. The mean width of the 90% posterior rank intervals was 36.50 positions under V0 and 36.74 under V1, with median width 37 under both models. Thus, simplifying V0 to the cross-classified V1 model changes little about either the broad ordering or the central finding that exact ranks are poorly identified. Detailed V0 ranking output and computational diagnostics are retained in Section 9.3.

4.3 Bayesian ranking using the equal-weight criterion average

The descriptive criterion analysis in Sections 2.4 and 2.5 suggests that the holistic grade and an equal-weight criterion average contain overlapping but not identical information. To examine whether this difference persists after accounting for reviewer-associated scoring tendencies and ranking uncertainty, V1 was fitted a second time using the review-level criterion average A_r as the response:

$$A_r \sim \mathcal{N}\left(\bar{A} + \theta_{p(r)}^{(A)} + b_{a(r)}^{(A)}, \sigma_A^2\right),$$

with

$$\theta_i^{(A)} \sim \mathcal{N}(0, \tau_{\theta, A}^2), \quad b_j^{(A)} \sim \mathcal{N}(0, \tau_{b, A}^2).$$

The reviewer–proposal assignments, priors and MCMC settings were kept identical to the overall-grade V1 fit; only the response and its observed mean changed. The resulting fit had satisfactory computational diagnostics (maximum PSRF 1.0032; minimum ESS 1,863).

Changing the response had a noticeably larger effect than changing from V0 to V1. The Spearman correlation between posterior-mean ranks from the two V1 fits was 0.873, and the Pearson correlation between expected ranks was 0.863. The mean absolute expected-rank change was 2.97 positions and the maximum was 7.89. By comparison, the Pearson correlation between V0 and V1 expected ranks using the same holistic outcome was 0.9916.

The expected-rank comparison is shown in Figure 4.

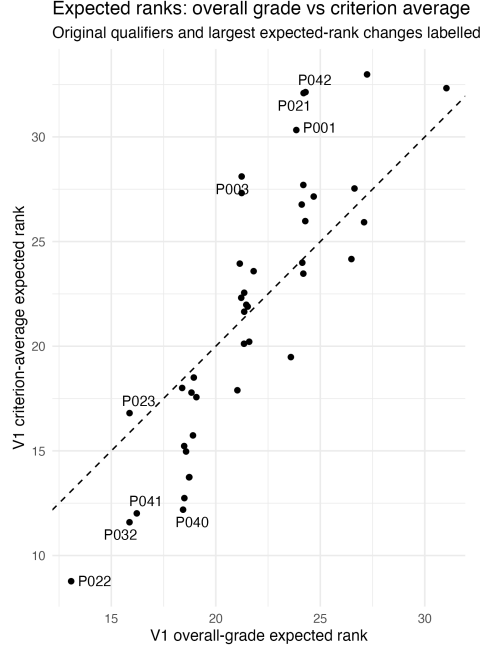


Figure 4: Posterior expected ranks from V1 fitted to the holistic overall grade and to the equal-weight criterion average. The dashed identity line represents identical expected ranks under both response definitions. CRS qualifiers and proposals with the largest expected-rank changes are labelled.

The upper ordering illustrates the difference. P022 and P032 remained at posterior-mean ranks 1 and 2 under the criterion-average model, P041 moved from rank 4 to rank 3, and P040 moved from rank 6 to rank 4. P023 moved from posterior-mean rank 3 to rank 11. That last change is less dramatic when uncertainty is considered: its expected rank changed only from 15.87 to 16.80, whereas P040 improved from 18.43 to 12.19. Expected-rank changes therefore provide a more informative comparison than exact integer ranks alone.

The criterion-average fit also produced somewhat narrower posterior rank distributions. The mean width of the 90% posterior rank intervals decreased from 36.74 to 33.19 positions, while the median width decreased from 37 to 34. The observed criterion average also had lower dispersion than the holistic grade, with standard deviations of 0.772 and 0.950, respectively. Since the criterion average combines four scores, part of this reduction in variation can arise mechanically from averaging. The corresponding change in the fitted variance decomposition is examined in Section 5.2.

The comparison does not establish that one score summary is preferable to the other. It shows that the definition of the evaluation outcome matters materially for the posterior ordering: changing from the holistic overall grade to the equal-

weight criterion average has a larger effect than changing from V0 to V1 while holding the holistic outcome fixed.

5 Validation and sensitivity analysis of V1

5.1 Validation strategy

The close agreement between V0 and V1 shows that the main ranking conclusion is not sensitive to the additional assessment-specific layer in V0, but it does not establish that V1 is an adequate working model. Validation therefore focuses on three questions: whether the posterior computation is stable, whether V1 reproduces important features of the observed grades, and whether the main conclusions depend materially on the baseline scale priors.

5.2 Computational and variance diagnostics

The V1 posterior was sampled reliably. Across all monitored parameters, the maximum PSRF was 1.0017 and the minimum effective sample size was 1,627. The corresponding trace plots are provided in Section 9.4.

To examine how V1 distributes variation across proposals, reviewers and individual reviews, note that its marginal variance is

$$\text{Var}(Y_r) = \tau_\theta^2 + \tau_b^2 + \sigma^2.$$

For posterior draw s , define

$$V^{(s)} = \left(\tau_\theta^{(s)}\right)^2 + \left(\tau_b^{(s)}\right)^2 + \left(\sigma^{(s)}\right)^2.$$

The corresponding variance shares are

$$\rho_{\text{proposal}}^{(s)} = \frac{\left(\tau_\theta^{(s)}\right)^2}{V^{(s)}}, \quad \rho_{\text{reviewer}}^{(s)} = \frac{\left(\tau_b^{(s)}\right)^2}{V^{(s)}}, \quad \rho_{\text{residual}}^{(s)} = \frac{\left(\sigma^{(s)}\right)^2}{V^{(s)}}.$$

For the overall-grade V1 model, the posterior medians of these shares were 0.050, 0.026 and 0.896, respectively. The corresponding 95% credible intervals were approximately [0.0002, 0.289], [0.0001, 0.240] and [0.615, 0.995].

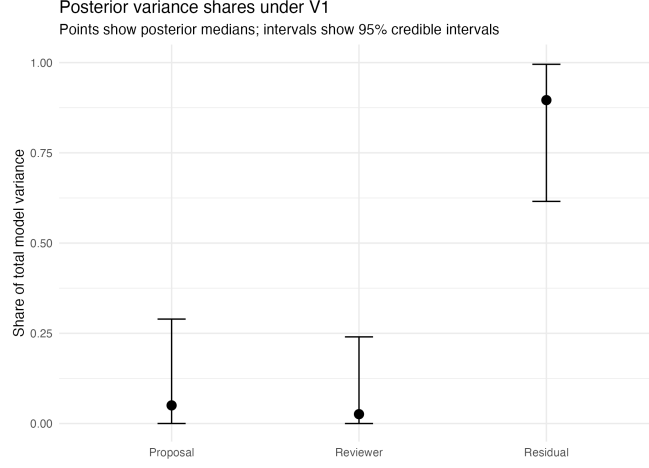


Figure 5: Posterior variance shares under V1. Points show posterior medians and vertical intervals show 95% credible intervals.

The proposal variance share also has a reliability interpretation. Consider two ratings of the same proposal i made by different reviewers j and k . Under V1,

$$Y_{ij} = \bar{Y} + \theta_i + b_j + \varepsilon_{ij}, \quad Y_{ik} = \bar{Y} + \theta_i + b_k + \varepsilon_{ik}.$$

The two ratings share the same proposal effect θ_i , whereas their reviewer effects and residual errors are distinct. Under the model assumptions that proposal effects, reviewer effects and residual errors are mutually independent, and that b_j and b_k are independent for $j \neq k$,

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Var}(\theta_i) = \tau_\theta^2.$$

Marginally, each rating has variance

$$\text{Var}(Y_{ij}) = \text{Var}(Y_{ik}) = \tau_\theta^2 + \tau_b^2 + \sigma^2.$$

It follows that

$$\text{Cor}(Y_{ij}, Y_{ik}) = \frac{\tau_\theta^2}{\tau_\theta^2 + \tau_b^2 + \sigma^2} = \rho_{\text{proposal}}.$$

Thus, under V1, the proposal variance share is also the model-implied intraproposal correlation: the expected correlation between two ratings of the same proposal by different reviewers. This corresponds to the cross-classified reliability interpretation used by Jayasinghe et al. (2003).

The posterior median of approximately 0.05 therefore indicates weak model-implied agreement attributable to a shared proposal effect. Most of the variation in individual grades is instead attributed to reviewer-specific and, predominantly, residual review-level variation. This helps explain why the posterior ranking remains broad despite observing two ratings per proposal, although the variance decomposition alone does not establish that the sparse assignment design is the cause of this uncertainty.

For comparison, the same variance decomposition was calculated for the V1 fit using the equal-weight criterion average A_r as the response.

Response	Proposal share	Reviewer share	Residual share
Holistic overall grade	0.050	0.026	0.896
Equal-weight criterion average	0.192	0.046	0.733

For every posterior draw, the three variance shares sum to one. The values reported in the table are the medians of the three posterior distributions calculated separately, so their reported medians need not sum exactly to one.

The criterion-average model attributes a larger share of its fitted variation to differences between proposals and a smaller share to residual review-level variation than the holistic-grade model. However, the two decompositions concern different response variables: A_r averages four criterion scores and has lower observed dispersion than the holistic grade. The larger proposal share should therefore not be interpreted as evidence that the equal-weight criterion average is intrinsically a better evaluation measure.

5.3 Posterior predictive assessment

Posterior predictive checks assessed whether V1 reproduces the observed location, dispersion and reviewer disagreement. For posterior draw s and review r ,

$$Y_r^{\text{rep},(s)} \sim \mathcal{N}\left(\bar{Y} + \theta_{p(r)}^{(s)} + b_{a(r)}^{(s)}, (\sigma^{(s)})^2\right).$$

The observed mean absolute disagreement between the two reviews of a proposal was 1.095, compared with a posterior predictive mean of 1.064. Approximately 40% of replicated datasets produced at least as much mean disagreement as observed, placing the observed value well within the central part of the posterior predictive distribution rather than in an extreme tail.

A more detailed check showed that the close correspondence is not mechanically induced by the shared proposal effect, which cancels when taking the difference between the two reviews of a proposal. The corresponding analytic expectation was 1.066, almost identical to the simulated value 1.064, and most of this disagreement was attributable to residual review-level variation rather than

persistent reviewer effects. Mapping replicated scores to the observed integer-valued 1–5 grid did not reduce the small difference in mean disagreement, but it substantially reduced the apparent discrepancy for disagreements of at least two score points, with the corresponding posterior predictive probability increasing from 0.0088 to 0.272. Full derivations and additional disagreement diagnostics are given in Section 9.5.

The main model-form limitation is visible in the lower-left panel of Figure 6. Because the Gaussian likelihood is unbounded, an average of 6.6% of replicated grades fall below 1 or above 5. V1 is therefore retained as a continuous approximation for latent proposal effects and ranking uncertainty, not as a literal generative model for valid future score categories.

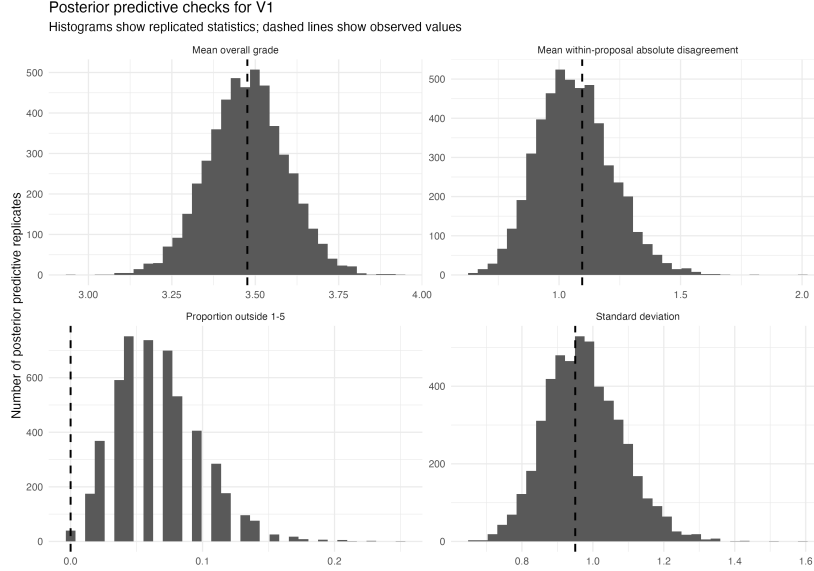


Figure 6: Posterior predictive checks for V1. Histograms show the distributions of statistics calculated from replicated datasets, while dashed vertical lines show the corresponding observed values.

5.4 Prior sensitivity

The baseline scale priors

$$\sigma, \tau_\theta, \tau_b \sim \text{Uniform}(0, 2)$$

were replaced by independent $\text{HalfNormal}(0, 1)$ priors while keeping the likelihood, data and MCMC settings unchanged. The ranking conclusions were essentially unchanged: the Pearson correlation between expected ranks from the

two prior specifications was 0.99994, the mean absolute expected-rank change was 0.093 positions and the maximum was 0.283. The median width of the 90% posterior rank intervals was 37 under both specifications. Posterior median variance shares were also nearly identical: the proposal share changed from 0.050 to 0.046, the reviewer share from 0.026 to 0.025 and the residual share from 0.896 to 0.902. The corresponding expected-rank figure and additional numerical details are provided in Section 9.7.

5.5 Validation conclusion

The validation provides no strong indication that the main V1 ranking conclusions are driven by computational instability, the tested scale priors or a failure to reproduce the observed mean, dispersion and reviewer-disagreement structure. The fitted model attributes most variation to residual review-level variation, while the exact proposal ranks remain weakly identified.

The principal model-form caveat is the continuous Gaussian treatment of the five-point score, which permits replicated values outside the admissible range. Subject to this approximation and the non-random reviewer assignment, V1 is retained as the working model. The substantial remaining uncertainty motivates the next chapter, which investigates whether the structure of the review design can identify a statistically informative single additional review.

6 Graph-based review design

6.1 Motivation and scope

The Bayesian analysis in the preceding chapters shows that the broad upper ordering of the CRS proposals is relatively stable, while the exact latent proposal ranks remain highly uncertain. Each proposal was evaluated by only two reviewers, and the reviewer-proposal assignments determine which proposal and reviewer effects are informed jointly by the observed grades.

The allocation of evaluations to reviewers is itself a statistical-design problem. In research peer review, Cook et al. (2005) formulate the allocation of proposals to reviewers with the explicit aim of facilitating an overall ranking, while Stelmakh, Shah and Singh (2021) study reviewer assignment under objectives involving fairness and statistical accuracy. Related work on statistical ranking goes one step further: Osting, Brune and Osher (2014) start from an existing pairwise-comparison graph and investigate how additional, targeted comparisons can increase the information available for ranking. In a Bayesian crowdsourcing setting, Simpson and Roberts (2015) similarly select informative agent-task assignments according to their expected information gain. These settings differ from CRS, but together they provide precedents for treating the allocation of additional evaluations as an information-design problem rather than as statistically neutral.

The purpose of the present chapter is not to assess whether the original CRS reviewer assignments were appropriate. Assignments were based on expertise and availability, and detailed information on reviewer eligibility, expertise and conflicts of interest is not available for the present analysis. Instead, the existing assignment structure is represented mathematically and its connection to posterior uncertainty under V1 is examined.

The analysis first considers a deliberately restricted prospective question: if exactly one additional review could hypothetically be obtained, which currently unobserved reviewer–proposal assignment would provide the greatest reduction in uncertainty under V1? Established optimal-design criteria are used rather than introducing a new measure of uncertainty, and several criteria are compared because different notions of statistical precision need not prefer the same additional review.

After establishing the single-review calculation, the primary A-optimality criterion is extended to a small multi-review budget. This provides a practical comparison between random allocation, targeting proposals with high observed reviewer disagreement, and sequential model-based allocation.

6.2 Reviewer–proposal assignment structure

The CRS assignment structure can be represented as a bipartite graph. A graph consists of vertices connected by edges, and it is called **bipartite** when the vertices can be separated into two groups such that every edge joins one group to the other.

This representation is closely related to the **Levi graph**, or incidence graph, used in the theory of incomplete block designs (Bailey and Cameron 2011). In a block design, one class of vertices represents treatments and the other represents blocks, with an edge recording that a treatment occurs in a block. For the present application, proposals play the role of treatments and reviewers play the role of blocks: a reviewer is connected to every proposal that the reviewer evaluated.

Formally, define

$$G = (P \cup R, E),$$

where

$$P = \{P_1, \dots, P_{42}\}$$

is the set of proposal vertices,

$$R = \{R_1, \dots, R_9\}$$

is the set of reviewer vertices, and

$$(P_i, R_j) \in E$$

whenever reviewer j evaluated proposal i .

The observed CRS design therefore has

$$|P| = 42, \quad |R| = 9, \quad |E| = 84.$$

The **degree** of a vertex is the number of edges incident to it. Every proposal vertex has degree two because each proposal received exactly two reviews. Reviewer degrees equal reviewer workloads and range from 6 to 12.

A small subset of the observed assignment graph is shown in Figure 7 to illustrate the bipartite structure. Reviewer vertices are placed on the left and proposal vertices on the right; an edge indicates that the reviewer evaluated the corresponding proposal. The figure is illustrative only and does not display the complete CRS assignment graph.

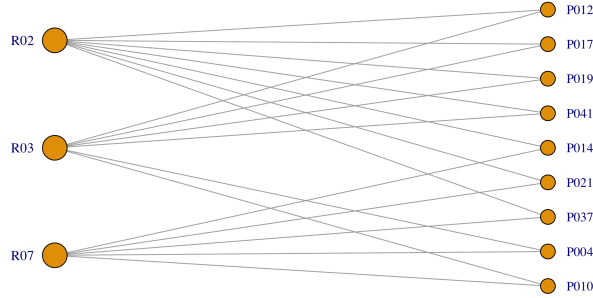


Figure 7: Illustrative subset of the observed CRS reviewer-proposal assignment graph. Reviewer vertices are shown on the left and proposal vertices on the right. Each edge represents one completed review. The subset is taken from the observed CRS assignment structure and is shown only to illustrate the bipartite representation used in the analysis.

The graph records only the assignment structure: an edge states that a review occurred, not which grade was assigned. It therefore separates the design of the

evaluation from the numerical outcomes subsequently observed.

6.3 Statistical relevance of the assignment structure

The graph representation is statistically relevant because the connection between incidence graphs and information matrices is well established in incomplete block-design theory. Bailey and Cameron (2011) consider an additive treatment–block model and show that, after changing the sign of the block parameters, the corresponding information structure can be represented through the Laplacian of the Levi graph. The CRS model has an analogous crossed structure, with proposal effects replacing treatment effects and reviewer effects replacing block effects.

To make this connection explicit, define

$$c_j = -b_j.$$

For an observed grade assigned by reviewer j to proposal i , $V1$ can then be written as

$$Y_{ij} - \bar{Y} = \theta_i - c_j + \varepsilon_{ij}.$$

Collect the proposal and transformed reviewer effects into

$$\beta = (\theta_1, \dots, \theta_{42}, c_1, \dots, c_9)^\top.$$

For each observed reviewer–proposal edge $e = (i, j)$, define an **oriented incidence vector** x_e containing +1 at the position corresponding to proposal i , −1 at the position corresponding to reviewer j , and zero elsewhere. Stacking the 84 incidence vectors gives an incidence matrix X , so that

$$Y - \bar{Y}\mathbf{1} = X\beta + \varepsilon.$$

For an oriented incidence matrix,

$$X^\top X = L_G,$$

where L_G is the graph Laplacian. Its diagonal entries are the vertex degrees, while an off-diagonal entry is −1 when the corresponding proposal and reviewer are connected and zero otherwise.

The same matrix enters the conditional posterior information in V1. Conditional on τ_θ , τ_b and σ , the prior precision for β is

$$P_0 = \text{diag}(\tau_\theta^{-2} I_{42}, \tau_b^{-2} I_9) = \begin{pmatrix} \tau_\theta^{-2} & & & & \\ & \ddots & & & \\ & & \tau_\theta^{-2} & & \\ & & & \tau_b^{-2} & \\ & & & & \ddots \end{pmatrix},$$

that is, the first 42 diagonal entries are τ_θ^{-2} and the remaining nine are τ_b^{-2} .

Combining prior and Gaussian likelihood gives conditional posterior precision

$$Q = P_0 + \frac{1}{\sigma^2} X^\top X = P_0 + \frac{1}{\sigma^2} L_G,$$

and conditional posterior covariance

$$V = Q^{-1}.$$

Thus, the assignment graph is not merely a visualization of the review structure: its Laplacian contributes directly to the conditional posterior precision of the proposal and reviewer effects. Changing an assignment changes L_G and hence changes the posterior covariance under the model. A full derivation of this matrix representation is given in Section 9.8.

6.4 Quantifying uncertainty for review design

The remaining question is how to define the statistical value of an additional review. Rather than constructing a new uncertainty measure for the CRS setting, the analysis uses established criteria from optimal experimental design. Bailey and Cameron (2011) distinguish A-, D- and E-optimality for block designs, corresponding respectively to average precision, the volume of a joint confidence region, and worst-case precision. Oosting, Brune and Osher (2014) use the same three optimality criteria in a statistical ranking problem in which the Fisher information is determined by a graph Laplacian. Their setting concerns direct pairwise comparisons rather than a reviewer-proposal graph, but it provides a direct precedent for using these criteria to assess the informativeness of additional edges in an existing ranking design.

Because the inferential target in the present analysis is the relative ordering of proposals, uncertainty is evaluated for **proposal contrasts** rather than for the absolute level of all model parameters. Let V_θ denote the 42×42 proposal-effect block of the conditional posterior covariance matrix V . For proposals i and k ,

$$\text{Var}(\theta_i - \theta_k \mid Y) = (e_i - e_k)^\top V_\theta (e_i - e_k),$$

where e_i denotes the i th standard basis vector.

The primary criterion used here is the **A-optimality criterion**, defined as the average posterior variance of all pairwise proposal contrasts,

$$L_A(V_\theta) = \frac{2}{42 \cdot 41} \sum_{i < k} (e_i - e_k)^\top V_\theta (e_i - e_k).$$

Smaller values therefore correspond to greater average precision when comparing two proposals. This criterion is particularly natural for the present ranking problem because proposal ordering depends directly on contrasts $\theta_i - \theta_k$. It is the Bayesian covariance analogue of the average-contrast criterion underlying A-optimality in classical block-design theory (Bailey and Cameron 2011).

Two alternative criteria are retained to assess whether the preferred additional review depends on the definition of uncertainty. Define the centering matrix

$$H = I_{42} - \frac{1}{42} \mathbf{1}\mathbf{1}^\top$$

where $\mathbf{1} \in \mathbb{R}^{42}$ denotes the vector of ones.

The matrix

$$HV_\theta H$$

represents posterior covariance on the proposal-contrast subspace. It has one zero eigenvalue corresponding to a common shift of all proposal effects and 41 non-zero eigenvalues, denoted $\lambda_1, \dots, \lambda_{41}$.

A **D-optimality** loss is defined by

$$L_D(V_\theta) = \sum_{\ell=1}^{41} \log(\lambda_\ell) = \log \text{pdet}(HV_\theta H),$$

where pdet denotes the **pseudo-determinant**, i.e. the product of the non-zero eigenvalues. Minimizing this quantity reduces the generalized volume of the joint posterior uncertainty region for proposal contrasts. It therefore provides a global criterion rather than averaging individual pairwise comparison variances.

Finally, an **E-optimality** loss is defined as

$$L_E(V_\theta) = \lambda_{\max}(HV_\theta H).$$

This criterion focuses on the least precisely estimated direction in the proposal-contrast space: minimizing it reduces the largest posterior variance among normalized contrast directions. In classical optimal-design terminology, this

is the covariance counterpart of maximizing the smallest relevant information eigenvalue (Bailey and Cameron 2011; Osting et al. 2014).

A-optimality is treated as the primary criterion because the substantive goal is comparison and ranking of proposals. D- and E-optimality are used as sensitivity criteria representing, respectively, global joint uncertainty and worst-case contrast uncertainty. The three quantities are therefore not new uncertainty measures introduced for this project; they are Bayesian posterior-covariance versions of established optimal-design objectives.

There is one methodological distinction from the classical block-design setting. Bailey and Cameron compare alternative designs as wholes, whereas the present CRS design has already been observed. The optimization is therefore restricted to **one-edge augmentations** of this fixed design. This restriction has a direct precedent in targeted data collection: Osting et al. (2014) augment an existing ranking graph with selected additional comparisons, while Simpson and Roberts (2015) provide a Bayesian precedent for choosing the next agent–task assignment according to its expected information value. In the present setting, every currently absent reviewer–proposal edge defines one hypothetical augmented design $G + e$. Section 6.5 compares these candidate designs by the reduction they produce in the three established criteria above.

6.5 Statistical value of a single additional review

The review-design calculation propagates uncertainty in the variance parameters rather than fixing them at a single posterior estimate. For each candidate assignment, the reduction in uncertainty is evaluated over 5,000 approximately evenly spaced retained posterior draws from V1 and then averaged across draws.

As a Monte Carlo stability check, the calculation was also performed using 1,000 posterior draws. The highest-ranked assignments and the relationships between the three design criteria were materially unchanged, so 5,000 draws were retained for the final analysis.

There are $42 \times 9 - 84 = 294$ currently unobserved reviewer–proposal pairs. Because reviewer expertise, eligibility and conflict-of-interest information is not available for this analysis, all 294 pairs are treated as hypothetical candidates. The resulting rankings therefore describe statistical informativeness under V1 rather than operational reviewer-assignment recommendations.

For posterior draw s , let

$$\tau_{\theta}^{(s)}, \quad \tau_b^{(s)}, \quad \sigma^{(s)}$$

denote the sampled proposal, reviewer and residual scale parameters. The corresponding prior precision matrix is

$$P_0^{(s)} = \text{diag} \left(\left(\tau_\theta^{(s)} \right)^{-2} I_{42}, \left(\tau_b^{(s)} \right)^{-2} I_9 \right).$$

Using the observed reviewer–proposal graph G , the conditional posterior precision and covariance matrices for that draw are

$$Q^{(s)} = P_0^{(s)} + \frac{1}{(\sigma^{(s)})^2} L_G,$$

and

$$V^{(s)} = (Q^{(s)})^{-1}.$$

Consider a currently unobserved reviewer–proposal assignment e and let x_e denote its oriented incidence vector, with $+1$ at the proposal vertex and -1 at the reviewer vertex. Adding this review changes the graph Laplacian by

$$L_{G+e} = L_G + x_e x_e^\top.$$

The corresponding conditional posterior covariance for draw s is

$$V_e^{(s)} = \left(Q^{(s)} + \frac{1}{(\sigma^{(s)})^2} x_e x_e^\top \right)^{-1}.$$

Because the additional review contributes a rank-one update to the precision matrix, the Sherman–Morrison identity gives

$$V_e^{(s)} = V^{(s)} - \frac{V^{(s)} x_e x_e^\top V^{(s)}}{(\sigma^{(s)})^2 + x_e^\top V^{(s)} x_e}.$$

Conditional on the variance parameters in draw s , the covariance reduction therefore depends only on which reviewer–proposal edge is added, not on the numerical grade that would eventually be observed.

For criterion $C \in \{A, D, E\}$, define the reduction in loss for candidate edge e and posterior draw s by

$$\Delta_C^{(s)}(e) = L_C(V_\theta^{(s)}) - L_C(V_{\theta,e}^{(s)}).$$

The overall value assigned to candidate edge e is the posterior average

$$\overline{\Delta}_C(e) = \frac{1}{S} \sum_{s=1}^S \Delta_C^{(s)}(e).$$

Larger values indicate a greater posterior-averaged reduction in uncertainty under the corresponding criterion. Averaging across posterior draws propagates the current uncertainty in τ_θ , τ_b and σ rather than replacing these parameters by single plug-in estimates.

As a sensitivity check, the calculation was repeated using posterior-median plug-in values for the three scale parameters instead of averaging utility over posterior draws. The resulting candidate rankings were largely preserved: the Spearman correlations with the posterior-averaged rankings were 0.873 for A-optimality, 0.990 for D-optimality and 0.894 for E-optimality. The five highest-ranked A-optimality candidates were identical as a set under both calculations, while four of the five highest-ranked candidates overlapped under D- and E-optimality. Because some lower-order candidate ranks changed, the posterior-averaged calculation is retained as the primary analysis. Additional details are provided in Section 9.9.

This calculation is not a full prospective refit of the hierarchical model. Within each posterior draw, the scale parameters are held fixed while the covariance change induced by the hypothetical additional review is evaluated. An actually observed additional grade could also update the posterior distribution of the scale parameters. The analysis therefore quantifies the information added by a new assignment conditional on the current V1 posterior, rather than modelling that subsequent learning step.

6.6 Comparison of single-review criteria

The 5,000-draw posterior-averaged calculation identified a small group of assignments with almost identical utility under A-optimality. The largest average reduction in pairwise proposal-contrast variance was obtained for P026–R02,

$$\overline{\Delta}_A = 3.0535 \times 10^{-4}.$$

P005–R02 and P015–R02 followed with $\overline{\Delta}_A = 3.0534 \times 10^{-4}$. The numerical differences between these three assignments are extremely small, so their exact ordering should not be interpreted as a meaningful separation. P032–R02, P029–R02 and P034–R02 formed the next tied group under A-optimality.

D-optimality gave a very similar upper ordering. P026–R02 was again ranked first, followed by P015–R02 and P005–R02. By contrast, E-optimality placed P005–R02 and P015–R02 first and second, but then preferred P005–R03 and P015–R03. The difference is summarized by the candidate-rank correlations,

$$\rho_S(A, D) = 0.926, \quad \rho_S(A, E) = 0.234, \quad \rho_S(D, E) = 0.040.$$

Thus, A- and D-optimality agree strongly about which additional assignments are informative, whereas E-optimality induces a substantially different ordering. This is consistent with the objectives defined in Section 6.4: A-optimality targets average pairwise contrast variance, D-optimality targets global joint uncertainty, and E-optimality targets the single least precise contrast direction.

Candidate assignment	A rank	D rank	E rank
P026–R02	1	1	18
P005–R02	2	3	1
P015–R02	3	2	2
P032–R02	4	4	5
P029–R02	4	5	6
P034–R02	4	5	7
P005–R03	8	41	3
P015–R03	9	42	4

The complete A-optimality pattern across all reviewer–proposal pairs is shown in Figure 8. Existing assignments are not candidates and are shown separately from the 294 possible additional edges. The highest A-optimality utilities are concentrated among assignments involving R02, but this should not be interpreted as evidence about reviewer quality. Conditional on the variance parameters, the design utility is determined by the assignment structure and the covariance model rather than by the future grade itself.

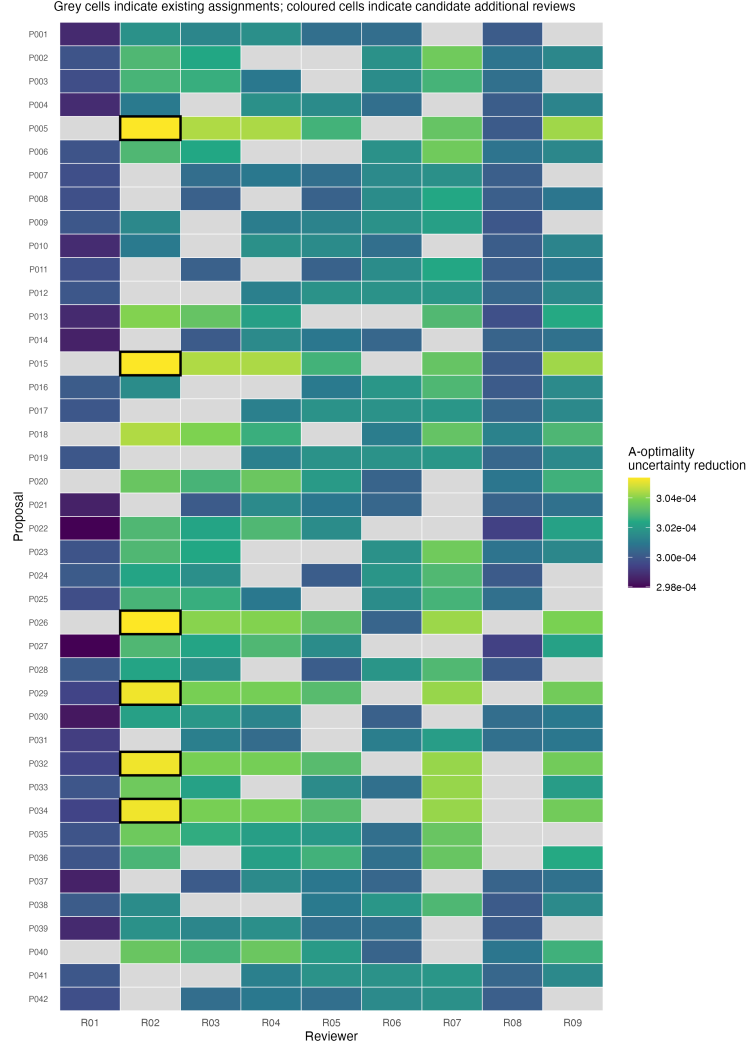


Figure 8: Posterior-averaged A-optimality utility for all hypothetical single additional reviewer–proposal assignments. Grey cells are assignments already observed and therefore unavailable as candidates. Outlined cells form the highest-ranked A-optimality group (ranks 1–4, including ties). Larger values indicate a greater reduction in average posterior pairwise proposal-contrast variance.

Targeting proposals with large observed reviewer disagreement answers a different question from A-optimal design. Such a rule prioritizes proposals for which the original reviewers expressed more divergent judgments, whereas A-optimality targets reduction in global posterior contrast uncertainty under V1. This distinction motivates using disagreement targeting as a deliberately different

practical benchmark in the following policy comparison rather than as another optimal-design criterion.

6.7 Practical comparison of limited-review allocation policies

The single-review analysis identifies which one additional evaluation would most reduce posterior uncertainty under a specified design criterion. In practice, however, an evaluation process may have a small budget of several additional reviews. To examine this setting, three allocation policies were compared for budgets of one to six additional reviews. Each proposal was allowed to receive at most one additional review, so the policies allocate selective third reviews rather than repeatedly reviewing the same proposal.

The **random policy** provides a no-targeting benchmark by selecting a currently unobserved reviewer–proposal assignment at random at each step.

The **disagreement-targeted policy** prioritizes proposals according to the absolute difference between their two original overall grades. Among proposals with the currently highest disagreement, the available reviewer–proposal edge with the largest A-optimality reduction is selected. Thus, observed disagreement determines *which proposal* receives another opinion, while A-optimality is used only to choose the most informative available reviewer within the targeted group.

The **sequential A-optimal policy** instead searches globally over all eligible remaining assignments. At each step, the edge with the largest posterior-averaged A-optimality reduction is selected and the conditional posterior covariance is updated before the next assignment is chosen. A-optimality is retained because it is the primary criterion in Section 6.4 and directly measures average uncertainty in pairwise proposal contrasts. The procedure is greedy and sequential; it is not claimed to be the jointly optimal set of several additional reviews.

For policy π and a budget of m additional reviews, define the posterior-averaged cumulative A-optimality reduction as

$$\bar{\Delta}_A(m; \pi) = \frac{1}{S} \sum_{s=1}^S \left[L_A(V_{\theta,0}^{(s)}) - L_A(V_{\theta,m,\pi}^{(s)}) \right],$$

where $V_{\theta,0}^{(s)}$ denotes the proposal covariance under the observed design and $V_{\theta,m,\pi}^{(s)}$ denotes the covariance after applying policy π for m additional reviews. To express this reduction relative to the initial level of uncertainty, define

$$G_m(\pi) = 100 \frac{\bar{\Delta}_A(m; \pi)}{\bar{L}_A(0)},$$

where

$$\bar{L}_A(0) = \frac{1}{S} \sum_{s=1}^S L_A(V_{\theta,0}^{(s)}).$$

The sequential policies were evaluated over 5,000 approximately evenly spaced V1 posterior draws. The random benchmark used 5,000 simulated allocation paths with a fixed random seed, balanced across the same posterior draws. The comparison therefore propagates the current uncertainty in the V1 scale parameters while using the same posterior resolution as the final single-review design analysis.

At a budget of six additional reviews, the posterior-averaged reduction in average pairwise proposal-contrast variance was 1.609% under sequential A-optimal allocation, 1.604% under disagreement-targeted allocation and 1.592% under random allocation. Thus, despite selecting substantially different reviewer–proposal assignments, the three policies produced very similar reductions in the global A-optimality criterion throughout the examined budget range.

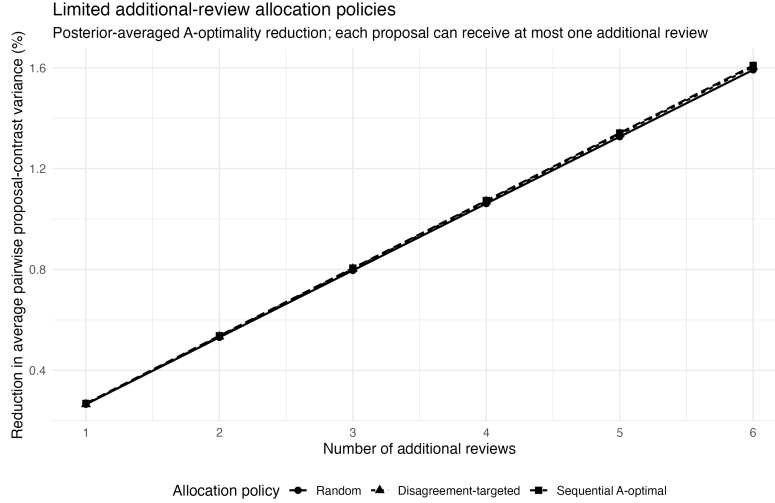


Figure 9: Posterior-averaged reduction in average pairwise proposal-contrast variance under random, disagreement-targeted and sequential A-optimal allocation of up to six additional reviews. Each proposal can receive at most one additional review. The near-overlap of the curves reflects the small difference in global A-optimality performance between the three allocation policies.

The near-equivalence is itself informative. Under V1 and the global A-optimality objective, optimizing the placement of a small number of additional reviews provides little improvement over random supplementation in the present CRS design. At a budget of six reviews, sequential A-optimal allocation exceeded the

random benchmark by only 0.017 percentage points. The disagreement-targeted policy nevertheless addresses a different substantive objective: it deliberately concentrates additional opinions on proposals for which the original reviewers disagreed. Its similar A-optimality performance therefore does not imply that disagreement targeting is unnecessary; rather, the policies pursue different allocation principles while producing very similar reductions in the global contrast-variance criterion considered here.

6.8 Generalization beyond the CRS design

The matrix formulation is not specific to the numerical CRS results. Any crossed reviewer–proposal model with an additive Gaussian information structure can represent observed assignments through an incidence matrix and evaluate how candidate edges change posterior covariance. The precise utility ranking and the relative performance of allocation policies will, however, depend on the fitted variance structure and on the uncertainty objective chosen.

The limited-budget comparison uses a greedy sequential rule: after each added edge, the information structure is updated and the next assignment is selected. This is computationally convenient but is not equivalent to jointly optimizing the complete set of additional reviews. The disagreement-targeted policy also uses only the disagreement observed in the original two reviews; it does not simulate a third grade and subsequently revise which proposals should be targeted.

A prospective implementation would first restrict the candidate set according to reviewer expertise, conflicts of interest, workload limits and availability. The statistical allocation rule would then operate only over feasible assignments. A fully adaptive Bayesian design could additionally observe each new grade, update the complete hierarchical model and then choose the next review.

7 Discussion and recommendations

7.1 Answers to the research questions

Research Question 1. Under V1, P022, P032, P023 and P041 occupied the first four posterior-mean positions. These proposals also had the most favourable broad ordering, but their exact ranks were weakly identified: the 90% posterior rank intervals extended to ranks 36–38. The posterior expected ranks of the four proposals ranged from 13.08 to 16.22, illustrating the strong movement of uncertain ranks toward the centre of the 42-proposal rank scale.

Research Question 2. The same four proposals that qualified under the final CRS rule occupied the first four posterior-mean V1 positions. V1 also closely agreed with the Heyard reference model V0, with Spearman correlation 0.995 between posterior-mean ranks and Pearson correlation 0.992 between expected ranks. The Bayesian analysis therefore did not overturn the observed upper

ordering. Its main contribution is to show that the fine ordering within and below that group is much less certain than an integer ranking suggests.

Research Question 3. V1 was computationally stable, with maximum PSRF 1.0017 and minimum effective sample size 1,627. Posterior predictive checks reproduced the observed mean, dispersion and mean reviewer disagreement reasonably well, and replacing the baseline scale priors by half-normal priors had negligible effect on the ranking. The main model-form caveat is the continuous Gaussian treatment of the integer-valued 1–5 score, which places about 6.6% of replicated grades outside the admissible range.

Research Question 4. Scientific quality was much more strongly associated with the holistic overall grade than the other three criteria, with Pearson and Spearman correlations of 0.892 and 0.890. An equal-weight average of all four criteria remained strongly related to the holistic grade but changed the Bayesian ordering materially: the correlation between V1 expected ranks under the two response definitions was 0.863, and only three of the four original qualifiers remained in the criterion-average posterior-mean top four. The analysis therefore shows that outcome definition matters, without establishing that either response is preferable.

Research Question 5. Under the primary A-optimality criterion, P026–R02, P005–R02 and P015–R02 formed an almost tied highest-value group, with P026–R02 ranked first by a very small numerical margin. A- and D-optimality gave strongly similar candidate rankings ($\rho_S = 0.926$), whereas E-optimality produced a substantially different ordering. These results identify statistically informative hypothetical assignments under V1; they are not operational recommendations because expertise, conflicts and eligibility are not included.

As a practical extension, random, disagreement-targeted and sequential A-optimal allocation were compared for up to six selective third reviews. Despite choosing different assignments, the three policies produced nearly identical reductions in the global A-optimality criterion. At six additional reviews, the corresponding reductions were 1.592%, 1.604% and 1.609%, respectively. Thus, under V1 and for the small review budgets considered here, the statistical benefit of optimizing the exact placement of additional reviews was small relative to the effect of adding reviews itself.

7.2 Interpretation of the main findings

The central finding is not a different winner produced by a Bayesian model. The upper ordering is stable across the observed scores, the final CRS qualification rule, V0 and V1, while the posterior distributions show that the exact latent ranks are highly uncertain. This is consistent with the fitted variance structure: the posterior median proposal share is only 0.050, whereas the residual share is 0.896. With two reviews per proposal, the data provide limited information for distinguishing many proposal effects precisely.

The analyses also separate two different sources of sensitivity. Simplifying the Heyard reference model from V0 to V1 changes the ranking very little, whereas replacing the holistic grade by an equal-weight criterion average changes it substantially more. The numerical ranking is therefore more sensitive to how the evaluation outcome is defined than to the distinction between the two tested hierarchical specifications.

The review-design analysis connects ranking uncertainty to the structure of data collection. Under V1, an additional assignment changes the graph Laplacian and therefore the conditional posterior covariance. The single-review analysis shows that individual assignments can be ranked according to their statistical information value and that the preferred edge depends on the uncertainty objective.

The limited-budget comparison adds an important qualification. Although sequential A-optimality, disagreement targeting and random allocation select different reviewer–proposal assignments, their reductions in global average pairwise contrast variance are almost indistinguishable for budgets of up to six additional reviews. In the present design, the precise placement of a small number of additional reviews therefore matters little for this particular global uncertainty objective. This does not preclude larger differences for decision-focused objectives or under different reviewer–proposal assignment structures.

7.3 Implications for CRS

For future CRS evaluations, the scoring rule used for the final decision should be specified as clearly as possible before reviews are collected. In the 2026 materials, the original call described an average across criteria and reviewers, while the implemented procedure used a separate holistic overall grade. The present analysis shows that these two response definitions can lead to materially different proposal orderings even when analysed with the same Bayesian model.

When only a small number of reviews are available, fine distinctions between proposal ranks should also be interpreted cautiously. Posterior rank intervals or selected pairwise posterior probabilities can complement a simple score table when the relative ordering itself matters. This does not require replacing a transparent qualification rule; it provides information about how strongly the available reviews support distinctions between proposals.

If additional review capacity is available, its allocation can nevertheless be treated as an explicit design choice. For the global A-optimality objective and the small review budgets considered here, however, model-based targeting provided little additional reduction in uncertainty relative to random supplementation. At six additional reviews, the reductions were 1.609% under sequential A-optimal allocation and 1.592% under random allocation. A complex optimization procedure would therefore not be justified solely by the global precision gain observed in this retrospective CRS design.

Other objectives may still motivate selective allocation. In particular, additional reviews could deliberately be directed toward proposals with strong reviewer disagreement or toward proposals close to a decision boundary. In a real CRS process, any statistical policy would first need to respect reviewer expertise, conflicts of interest, availability and workload. The anonymized reviewer IDs identified here should therefore not be interpreted as direct assignment recommendations.

7.4 Ranking uncertainty and decision making

The final CRS rule and the Bayesian ranking answer different questions. The CRS rule is deterministic once the two observed grades are known: in 2026 a proposal qualified if it received at least one grade of 5 and no grade below 4. The Bayesian model instead estimates latent proposal effects and asks how uncertain their relative ordering remains after accounting for reviewer effects and residual variation.

Consequently, broad posterior rank intervals do not imply that the four funding decisions were incorrect. All four observed qualifiers were also the first four proposals by posterior-mean V1 effect, and the available capacity was not binding. The uncertainty result is narrower: the data do not support strong claims about the exact latent ordering of many proposals. If a future decision required a strict ordering near a funding boundary, this distinction between observed qualification and latent rank uncertainty would become more important.

7.5 Limitations

The first limitation is the size and sparsity of the dataset. The analysis is based on 42 proposals, nine reviewers and 84 reviews, with exactly two reviews per proposal. Variance components and proposal ranks are therefore estimated from limited replication, and the resulting uncertainty is itself an important part of the result.

Second, reviewer assignments were based on expertise and availability rather than random allocation. Reviewer effects in V1 consequently describe reviewer-associated scoring tendencies and should not be interpreted causally as reviewer bias. The same issue limits the Chapter 6 design analysis: expertise, conflicts of interest, workload constraints and availability are absent from the candidate set.

Third, the review-design calculations condition on the current V1 posterior and evaluate covariance updates with fixed scale parameters within each draw. The sequential policy comparison repeatedly updates this conditional covariance structure, but it does not simulate the numerical grades that would be observed or re-fit the full hierarchy after each additional review. The disagreement-targeted policy also uses disagreement in the original two reviews rather than updating the targeting rule after observing a third score.

Fourth, V1 treats the five-point holistic grade as Gaussian with homogeneous residual variance. This provides a simple continuous ranking model but allows

replicated scores outside the observed 1–5 range. The prior-sensitivity analysis does not address this likelihood assumption, and an ordinal model was not fitted as a formal sensitivity analysis.

Fifth, the criterion-average analysis is an equal-weight counterfactual. It does not recover the weights reviewers implicitly used when assigning the holistic grade and does not establish that equal weighting would be a better decision rule.

Finally, the study is a retrospective analysis of one small grant call. The qualitative conclusions about sparse review, uncertainty and targeted additional information may transfer to related settings, but the numerical rankings and preferred review assignments are specific to the CRS design and model fitted here.

7.6 Future work

A natural model extension is an ordinal hierarchical specification that respects the five observed score categories while retaining proposal and reviewer effects. Comparing such a model with V1 would directly assess whether the continuous-score approximation affects the ranking conclusions.

For review design, the next step is to incorporate operational eligibility constraints and compare the present greedy sequential rule with joint optimization under a fixed review budget. A fully adaptive prospective design could observe each additional grade, update the complete Bayesian model and only then choose the next assignment. Decision-focused utilities could also be considered when the primary goal is resolving uncertainty for particular proposals, such as cases with substantial reviewer disagreement or proposals near a qualification boundary, rather than reducing global ranking uncertainty.

The limited-budget comparison also raises a broader structural question: under which review designs does graph-based targeting provide a meaningful advantage over random supplementation? One hypothesis is that the advantage increases when reviewer–proposal graphs contain stronger workload imbalance, clustering or connectivity bottlenecks, particularly when persistent reviewer variation is large relative to residual review-level variation. In such settings, an additional edge that connects weakly linked parts of the assignment graph may carry substantially more information than another review within an already well-connected region. A simulation study could vary both the assignment-graph structure and the variance components systematically to test this hypothesis. Targeting may also become more consequential when the uncertainty criterion is restricted to a small decision-relevant subset of proposals rather than averaged over all proposal pairs.

The strongest empirical validation would be prospective. Applying a pre-specified ranking and review-design procedure in a future CRS call would allow the predicted information gains to be compared with the uncertainty actually observed

after additional reviews are collected.

8 Conclusion

The CRS Seed Grant 2026 evaluation provides a small but informative example of the uncertainty created when many proposals are assessed by only a few reviewers. A Bayesian cross-classified model preserved the same four strongest posterior-mean proposals as the final CRS qualification procedure, while showing that the exact latent ranking remained highly uncertain. The main ranking conclusion was stable to the more complex Heyard reference specification and to the tested scale priors, although the continuous Gaussian treatment of the five-point score remains a modelling limitation.

The analysis also showed that the choice of evaluation outcome matters. An equal-weight average of the four supporting criteria produced a more noticeable change in proposal ordering than the change from V0 to V1. This supports specifying clearly in advance whether a holistic grade, criterion-based score or another summary is intended to drive the decision.

Finally, the reviewer–proposal assignment graph provides a direct connection between the review design and posterior uncertainty under V1. A-, D- and E-optimality can be used to evaluate the statistical value of individual additional reviews, but the limited-budget comparison showed that random, disagreement-targeted and sequential A-optimal allocation produced very similar reductions in global average pairwise contrast variance. Thus, while the graph framework provides a principled way to evaluate possible assignments, complex targeting was not practically important for this objective in the present CRS design. The broader practical takeaway is to combine transparent decision rules with explicit uncertainty summaries and to define clearly what additional review effort is intended to resolve before choosing an allocation strategy.

9 Supplementary methods and analyses

9.1 Correlation definitions

For criterion c , the sample Pearson correlation with the holistic overall grade is

$$\text{Cor}_P(C_{\cdot c}, Y) = \frac{\sum_{r=1}^n (C_{rc} - \bar{C}_c)(Y_r - \bar{Y})}{\sqrt{\sum_{r=1}^n (C_{rc} - \bar{C}_c)^2 \sum_{r=1}^n (Y_r - \bar{Y})^2}},$$

where $n = 84$, $C_{\cdot c} = (C_{1c}, \dots, C_{nc})$ and $Y = (Y_1, \dots, Y_n)$. The corresponding sample Spearman correlation is

$$\text{Cor}_S(C_{\cdot c}, Y) = \text{Cor}_P(R(C_{\cdot c}), R(Y)),$$

where $R(\cdot)$ denotes the rank transformation and average ranks are used for ties.

9.2 Bayesian updating and partial pooling in V1

This section gives a short mathematical explanation of how the observed grades inform the proposal and reviewer effects in V1. The purpose is to make the Bayesian updating mechanism and the resulting partial pooling explicit.

9.2.1 From the model to the posterior distribution

Recall that V1 represents review r as

$$Y_r = \bar{Y} + \theta_{p(r)} + b_{a(r)} + \varepsilon_r, \quad \varepsilon_r \sim \mathcal{N}(0, \sigma^2),$$

where θ_i is the latent effect of proposal i , b_j is the effect of reviewer j , and σ^2 is the residual review-level variance. The hierarchical distributions are

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2), \quad b_j \sim \mathcal{N}(0, \tau_b^2).$$

For given values of the model parameters, the density of one observed grade is

$$p(Y_r \mid \theta, b, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(Y_r - \bar{Y} - \theta_{p(r)} - b_{a(r)})^2}{2\sigma^2} \right].$$

Conditional on the model parameters, the review-level residuals are assumed independent. The likelihood of the complete observed dataset $Y = (Y_1, \dots, Y_{84})$ is therefore

$$p(Y \mid \theta, b, \sigma) = \prod_{r=1}^{84} p(Y_r \mid \theta, b, \sigma).$$

Bayes' theorem combines this likelihood with the prior distributions. For all unknown quantities together,

$$p(\theta, b, \tau_\theta, \tau_b, \sigma \mid Y) = \frac{p(Y \mid \theta, b, \sigma) p(\theta, b, \tau_\theta, \tau_b, \sigma)}{p(Y)}.$$

Since the denominator $p(Y)$ does not depend on the unknown parameters, the same relationship can be written as

$$p(\theta, b, \tau_\theta, \tau_b, \sigma \mid Y) \propto p(Y \mid \theta, b, \sigma) p(\theta, b, \tau_\theta, \tau_b, \sigma),$$

where $\propto$ means “proportional to”. Using the hierarchical structure of V1 gives

$$p(\theta, b, \tau_\theta, \tau_b, \sigma \mid Y) \propto p(Y \mid \theta, b, \sigma) \left[\prod_{i=1}^{42} p(\theta_i \mid \tau_\theta) \right] \left[\prod_{j=1}^9 p(b_j \mid \tau_b) \right] p(\tau_\theta) p(\tau_b) p(\sigma).$$

The posterior therefore combines two sources of information. The likelihood describes how compatible particular parameter values are with the observed grades, while the priors describe how plausible those parameter values are under the hierarchical model before conditioning on the observed grades.

9.2.2 Updating one proposal effect

The partial-pooling mechanism can be seen explicitly by considering one proposal i . Let

$$\mathcal{R}_i = \{r : p(r) = i\}$$

denote the set of reviews received by proposal i , and define

$$n_i = |\mathcal{R}_i|.$$

Thus, n_i is simply the number of reviews received by proposal i . In the CRS data,

$$n_i = 2$$

for every proposal.

To make the updating calculation transparent, suppose temporarily that the reviewer effects b , the residual variance σ^2 , and the between-proposal variance τ_θ^2 are known.

For a review $r \in \mathcal{R}_i$, define the reviewer-adjusted grade

$$Z_r = Y_r - \bar{Y} - b_{a(r)}.$$

Substituting the V1 model gives

$$Z_r = \theta_i + \varepsilon_r, \quad \varepsilon_r \sim \mathcal{N}(0, \sigma^2),$$

and therefore

$$Z_r \mid \theta_i \sim \mathcal{N}(\theta_i, \sigma^2).$$

The likelihood for θ_i based on its n_i adjusted grades is

$$p(Z_i \mid \theta_i, \sigma) = \prod_{r \in \mathcal{R}_i} p(Z_r \mid \theta_i, \sigma),$$

where Z_i denotes the collection of adjusted grades for proposal i . Ignoring multiplicative constants that do not depend on θ_i ,

$$p(Z_i \mid \theta_i, \sigma) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{r \in \mathcal{R}_i} (Z_r - \theta_i)^2 \right].$$

The prior distribution

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2)$$

has density, again ignoring constants independent of θ_i ,

$$p(\theta_i \mid \tau_\theta) \propto \exp \left(-\frac{\theta_i^2}{2\tau_\theta^2} \right).$$

Bayes' theorem gives

$$p(\theta_i \mid Z_i, \sigma, \tau_\theta) \propto p(Z_i \mid \theta_i, \sigma) p(\theta_i \mid \tau_\theta),$$

so

$$p(\theta_i \mid Z_i, \sigma, \tau_\theta) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{r \in \mathcal{R}_i} (Z_r - \theta_i)^2 - \frac{\theta_i^2}{2\tau_\theta^2} \right].$$

To identify this posterior distribution, expand the square:

$$\sum_{r \in \mathcal{R}_i} (Z_r - \theta_i)^2 = \sum_{r \in \mathcal{R}_i} Z_r^2 - 2\theta_i \sum_{r \in \mathcal{R}_i} Z_r + n_i \theta_i^2.$$

The term $\sum Z_r^2$ does not depend on θ_i and can therefore be absorbed into the proportionality constant. The remaining terms give

$$p(\theta_i \mid Z_i, \sigma, \tau_\theta) \propto \exp \left[-\frac{1}{2} \left\{ \left(\frac{n_i}{\sigma^2} + \frac{1}{\tau_\theta^2} \right) \theta_i^2 - 2 \left(\frac{\sum_{r \in \mathcal{R}_i} Z_r}{\sigma^2} \right) \theta_i \right\} \right].$$

A normal density with mean $\mu_i^{(\theta)}$ and variance $v_i^{(\theta)}$ has the same quadratic form,

$$\exp \left[-\frac{1}{2v_i^{(\theta)}} \left(\theta_i - \mu_i^{(\theta)} \right)^2 \right].$$

Expanding this square and comparing coefficients gives

$$\frac{1}{v_i^{(\theta)}} = \frac{1}{\tau_\theta^2} + \frac{n_i}{\sigma^2},$$

and therefore

$$v_i^{(\theta)} = \left(\frac{1}{\tau_\theta^2} + \frac{n_i}{\sigma^2} \right)^{-1}.$$

The coefficient of θ_i gives

$$\frac{\mu_i^{(\theta)}}{v_i^{(\theta)}} = \frac{\sum_{r \in \mathcal{R}_i} Z_r}{\sigma^2},$$

so

$$\mu_i^{(\theta)} = v_i^{(\theta)} \frac{\sum_{r \in \mathcal{R}_i} Z_r}{\sigma^2}.$$

Now define the mean reviewer-adjusted grade for proposal i as

$$\bar{Z}_i = \frac{1}{n_i} \sum_{r \in \mathcal{R}_i} Z_r.$$

Since

$$\sum_{r \in \mathcal{R}_i} Z_r = n_i \bar{Z}_i,$$

the posterior mean simplifies to

$$\mu_i^{(\theta)} = \frac{n_i \tau_\theta^2}{\sigma^2 + n_i \tau_\theta^2} \bar{Z}_i.$$

Thus,

$$\theta_i \mid Z_i, \sigma, \tau_\theta \sim \mathcal{N} \left(\mu_i^{(\theta)}, v_i^{(\theta)} \right).$$

9.2.3 Partial pooling

Define the shrinkage factor

$$\kappa_i = \frac{n_i \tau_\theta^2}{\sigma^2 + n_i \tau_\theta^2}.$$

Because all variance terms are positive,

$$0 < \kappa_i < 1.$$

The conditional posterior mean can therefore be written as

$$\mu_i^{(\theta)} = \kappa_i \bar{Z}_i.$$

The reviewer-adjusted observed mean $\bar{Z}_i$ is consequently pulled toward the prior mean zero rather than being used directly as the estimate of the proposal effect. This is the partial-pooling mechanism.

The amount of pooling follows directly from κ_i . More reviews increase n_i and therefore give the observed grades more influence. Larger residual variation σ^2 decreases κ_i and produces stronger shrinkage, because individual reviews are treated as noisier. Larger between-proposal variation τ_θ^2 increases κ_i , allowing proposal effects to remain further away from zero.

For the CRS design, $n_i = 2$ for every proposal, so

$$\kappa_i = \frac{2\tau_\theta^2}{\sigma^2 + 2\tau_\theta^2}.$$

Since $\theta_i = 0$ corresponds to a proposal whose expected grade equals the overall mean $\bar{Y}$, shrinking θ_i toward zero corresponds to shrinking the proposal's fitted grade toward the overall evaluation mean.

9.2.4 Updating reviewer effects

Reviewer effects are updated in the same way. Let

$$\mathcal{S}_j = \{r : a(r) = j\}$$

denote the set of reviews completed by reviewer j , and define

$$d_j = |\mathcal{S}_j|.$$

Thus, d_j is the number of reviews completed by reviewer j .

Suppose temporarily that the proposal effects θ , the residual variance σ^2 , and the between-reviewer variance τ_b^2 are known. Define the proposal-adjusted grade

$$W_r = Y_r - \bar{Y} - \theta_{p(r)}.$$

For reviews completed by reviewer j ,

$$W_r = b_j + \varepsilon_r,$$

so

$$W_r \mid b_j \sim \mathcal{N}(b_j, \sigma^2).$$

Combining these observations with the prior

$$b_j \sim \mathcal{N}(0, \tau_b^2)$$

and applying the same normal-normal calculation gives

$$b_j \mid Y, \theta, \sigma, \tau_b \sim \mathcal{N}(\mu_j^{(b)}, v_j^{(b)}),$$

where

$$v_j^{(b)} = \left(\frac{1}{\tau_b^2} + \frac{d_j}{\sigma^2} \right)^{-1}.$$

Define the mean proposal-adjusted grade assigned by reviewer j as

$$\bar{W}_j = \frac{1}{d_j} \sum_{r \in \mathcal{S}_j} W_r.$$

The conditional posterior mean is then

$$\mu_j^{(b)} = \frac{d_j \tau_b^2}{\sigma^2 + d_j \tau_b^2} \bar{W}_j.$$

Thus, the several proposals assessed by the same reviewer jointly inform that reviewer's scoring tendency. Reviewers with more observed reviews are, all else equal, subject to less shrinkage toward zero.

9.2.5 From the joint posterior to proposal ranks

The calculations above treat some quantities as known only to make the updating mechanism visible. In the fitted V1 model, the proposal effects, reviewer effects, τ_θ , τ_b , and σ are all unknown and are estimated jointly. Their posterior distributions therefore depend on one another.

JAGS uses Markov chain Monte Carlo to generate draws from this joint posterior. A retained posterior draw s contains

$$\left(\theta_1^{(s)}, \dots, \theta_{42}^{(s)}, b_1^{(s)}, \dots, b_9^{(s)}, \tau_\theta^{(s)}, \tau_b^{(s)}, \sigma^{(s)}\right).$$

Each draw represents one set of parameter values that is compatible with the observed grades and the model assumptions. The proposal effects within that draw,

$$\theta_1^{(s)}, \dots, \theta_{42}^{(s)},$$

are ordered from largest to smallest to obtain one possible proposal ranking. Repeating this calculation across posterior draws produces the posterior rank distributions described in Section 3.5.

The Bayesian ranking therefore follows the sequence

observed grades $\longrightarrow$ posterior proposal and reviewer effects $\longrightarrow$ posterior proposal ranks.

Partial pooling enters before the ranking step: the limited and noisy review information is combined with the hierarchical model before the proposal effects are ranked.

9.3 V0 reference-model details

Under V0, P022 had posterior-mean rank 1 and expected rank 11.85. P023, P032 and P041 occupied posterior-mean ranks 2, 3 and 4, with expected ranks 15.45, 15.49 and 16.50. Their 90% posterior rank intervals were 1–35, 1–37, 1–37 and 2–38, respectively. The maximum PSRF across monitored V0 parameters was 1.0077 and the minimum effective sample size was 501.

For completeness, Figure 10 shows the comparison between the observed-score rank, posterior-mean rank and expected rank under V0.

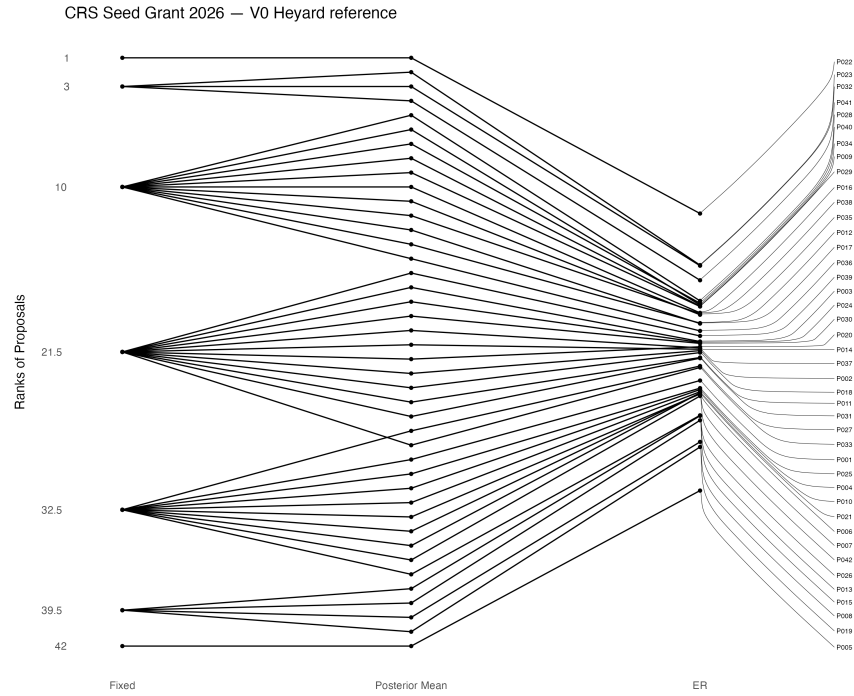


Figure 10: Proposal rankings under the observed-score ranking, posterior-mean ranking and posterior expected rank for the V0 Heyard reference model.

9.4 V1 MCMC trace plots

Figure 11 shows the final 5,000 stored iterations from each V1 chain, plotting every fifth value for readability. Numerical diagnostics in the main text are based on the full posterior sample.

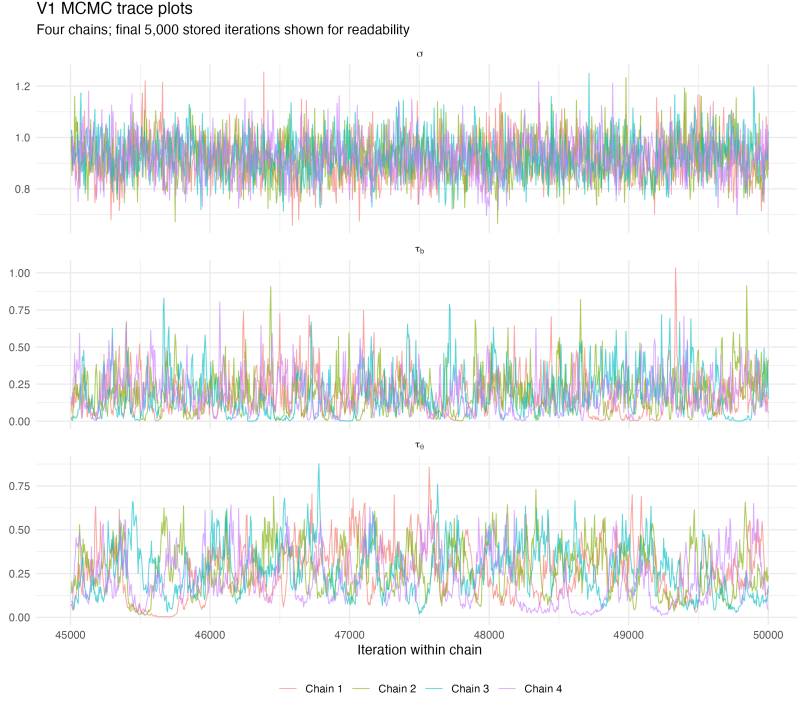


Figure 11: Trace plots for the V1 scale parameters τ_θ , τ_b , and σ .

9.5 Detailed reviewer-disagreement diagnostic

Consider proposal i , reviewed by reviewers j and k . Under V1,

$$Y_{ij} - Y_{ik} = b_j - b_k + \varepsilon_{ij} - \varepsilon_{ik},$$

so the proposal effect cancels and

$$Y_{ij} - Y_{ik} \mid b_j, b_k, \sigma \sim \mathcal{N}(b_j - b_k, 2\sigma^2).$$

Taking the absolute value gives a folded normal distribution ([Tsagris et al. 2014](#)). If $X \sim \mathcal{N}(\mu, s^2)$, then

$$\mathbb{E}|X| = s\sqrt{\frac{2}{\pi}} \exp\left(-\frac{\mu^2}{2s^2}\right) + |\mu| \left[2\Phi\left(\frac{|\mu|}{s}\right) - 1\right],$$

where Φ is the standard normal cumulative distribution function. In the present setting, $\mu = b_j - b_k$ and $s = \sqrt{2}\sigma$. Evaluating this expression for the observed reviewer pairs and posterior draws gave analytic posterior mean disagreement

1.066, almost identical to the simulated posterior predictive mean 1.064. The posterior predictive probability of mean disagreement at least as large as the observed value 1.095 was 0.401.

Most of the model-implied disagreement is associated with residual review-level variation. If the two reviewer effects are equal, the expected absolute disagreement simplifies to

$$E|Y_{ij} - Y_{ik}| = \frac{2\sigma}{\sqrt{\pi}},$$

which averaged 1.043 over the posterior. Allowing the fitted reviewer effects to differ increased the expected mean disagreement by only about 0.023 on average.

A threshold-based check initially suggested underprediction of moderate disagreement. Observed absolute disagreement was 0 for 10 proposals, 1 for 19 proposals, 2 for 12 proposals and 3 for one proposal. Thus 13 of 42 proposals had disagreement of at least two score points, giving

$$T_{\text{obs}} = \frac{13}{42} \approx 0.310.$$

With continuous Gaussian replicated grades, the posterior predictive mean of the corresponding proportion was 0.134 and

$$\mathbb{P}(T_{\text{cont}}^{\text{rep}} \geq T_{\text{obs}} \mid Y) = 0.0088.$$

Because observed scores are integer-valued, a diagnostic version of each replicated score was mapped to the 1–5 grid,

$$\tilde{Y}_r^{\text{rep}} = \min \{5, \max \{1, \text{round}(Y_r^{\text{rep}})\}\}.$$

After this transformation, the posterior predictive mean proportion with disagreement of at least two increased to 0.255 and

$$\mathbb{P}(T_{\text{disc}}^{\text{rep}} \geq T_{\text{obs}} \mid Y) = 0.272.$$

The same transformation did not improve the agreement for the mean absolute disagreement. The posterior predictive mean changed from 1.064 under the continuous Gaussian replication to 1.020 after mapping the replicated scores to the 1–5 grid, compared with the observed value 1.095. The corresponding 95% posterior predictive interval for the discretized mean disagreement was approximately [0.738, 1.333], and

$$\mathbb{P}(D_{\text{disc}}^{\text{rep}} \geq D_{\text{obs}} \mid Y) = 0.332.$$

Thus, the grid transformation primarily resolves the apparent discrepancy in the threshold-based disagreement statistic rather than the already small difference in mean absolute disagreement.

The broader disagreement pattern is summarized in Figure 12. The observed mean, median, 90th percentile, maximum and proportion of disagreements of at least three score points are all compatible with their posterior predictive distributions. The clearest departure is the frequency of disagreements of at least two score points, which is higher in the observed data than in most continuous replicated datasets. As shown above, this discrepancy is substantially reduced once replicated grades are placed on the same integer-valued 1–5 scale as the observations.

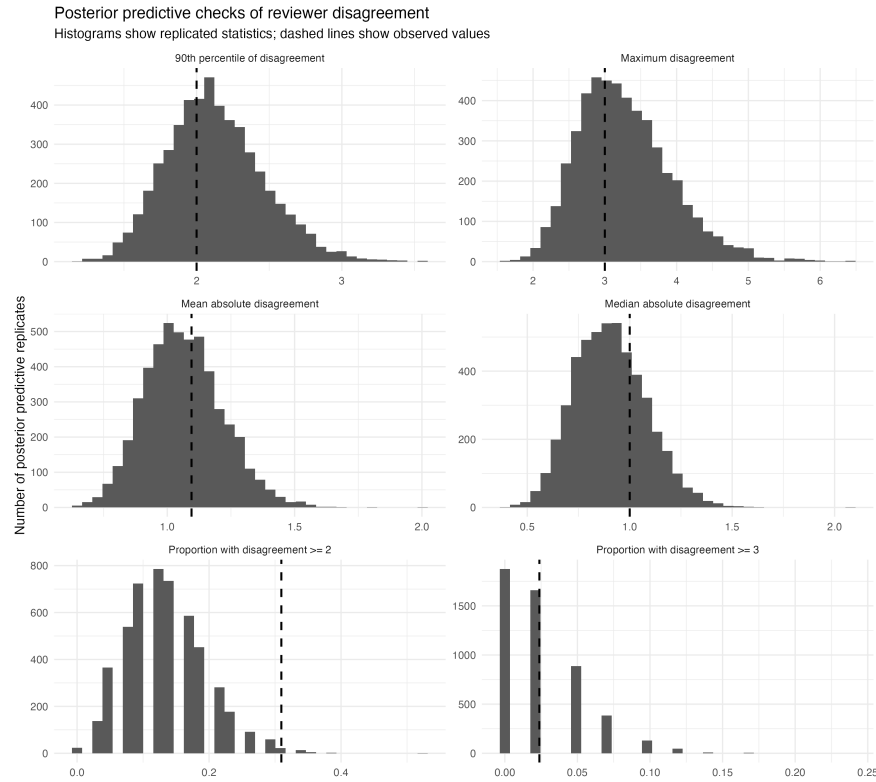


Figure 12: Posterior predictive checks for several summaries of reviewer disagreement under V1. Histograms show statistics from replicated datasets and dashed vertical lines show the corresponding observed values.

9.6 Additional posterior predictive summaries

Statistic	Observed	Replicated mean	2.5%	97.5%
Mean overall grade	3.476	3.474	3.241	3.704
Standard deviation	0.950	0.975	0.781	1.201
Mean absolute disagreement	1.095	1.064	0.790	1.386
Proportion outside 1–5	0.000	0.066	0.012	0.143

The intervals are the 2.5th and 97.5th percentiles of the corresponding statistics across the 5,000 posterior predictive datasets.

9.7 Prior-sensitivity details

Under the HalfNormal(0, 1) scale priors, the posterior medians of τ_θ , τ_b and σ were 0.209, 0.154 and 0.916, compared with 0.220, 0.157 and 0.920 under the baseline specification. The maximum PSRF was 1.0011 and the minimum effective sample size was 1,306.

The Spearman correlation between posterior-mean ranks under the two prior specifications was 0.9990. Six proposals changed integer posterior-mean rank, but the changes were small: the mean absolute change was 0.19 positions and the maximum was two positions. The mean widths of the 90% posterior rank intervals were 36.74 and 36.83 under the baseline and half-normal specifications, respectively.

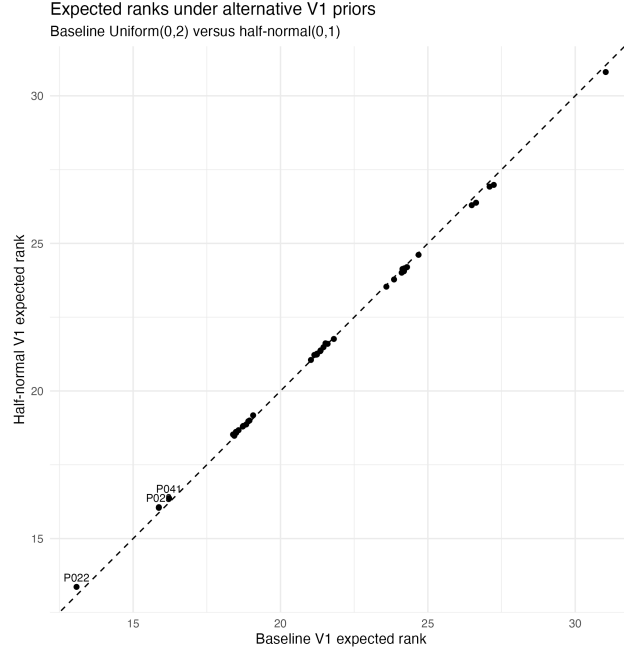


Figure 13: Posterior expected ranks under the baseline V1 prior specification and the half-normal prior-sensitivity specification. The dashed identity line represents identical expected ranks under both models.

9.8 Graph representation and conditional posterior precision

This section derives the connection between the reviewer–proposal assignment graph and the conditional posterior uncertainty in V1. The construction is closely related to the use of incidence graphs and graph Laplacians in incomplete block-design theory ([Bailey and Cameron 2011](#)), but is written here for the Bayesian cross-classified model used in this report.

Recall that V1 is

$$Y_r = \bar{Y} + \theta_{p(r)} + b_{a(r)} + \varepsilon_r, \quad \varepsilon_r \sim \mathcal{N}(0, \sigma^2).$$

To obtain the usual signed incidence-matrix representation, define

$$c_j = -b_j.$$

The model can then be written as

$$Y_r - \bar{Y} = \theta_{p(r)} - c_{a(r)} + \varepsilon_r.$$

Collect the 42 proposal effects and nine transformed reviewer effects into the 51-dimensional vector

$$\beta = (\theta_1, \dots, \theta_{42}, c_1, \dots, c_9)^\top.$$

For every observed review r , corresponding to an edge between proposal $p(r)$ and reviewer $a(r)$, define a row vector $x_r^\top \in \mathbb{R}^{51}$ with +1 in the position corresponding to proposal $p(r)$, -1 in the position corresponding to reviewer $a(r)$, and zero elsewhere. Stacking the 84 row vectors gives the oriented incidence matrix

$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_{84}^\top \end{pmatrix},$$

which has dimension 84×51 .

Define the centered response vector

$$y = Y - \bar{Y}\mathbf{1}.$$

The complete V1 model can then be written in matrix form as

$$y = X\beta + \varepsilon,$$

where

$$\varepsilon \sim \mathcal{N}(0, \sigma^2 I_{84}).$$

Conditional on σ , the likelihood for β is therefore

$$p(y \mid \beta, \sigma) \propto \exp \left[-\frac{1}{2\sigma^2} (y - X\beta)^\top (y - X\beta) \right].$$

Expanding the quadratic term gives

$$(y - X\beta)^\top (y - X\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

Since $y^\top y$ does not depend on β , the part of the likelihood involving β is

$$p(y \mid \beta, \sigma) \propto \exp \left[-\frac{1}{2} \left\{ \beta^\top \left(\frac{1}{\sigma^2} X^\top X \right) \beta - 2\beta^\top \left(\frac{1}{\sigma^2} X^\top y \right) \right\} \right].$$

Thus, the likelihood contributes the precision matrix

$$\frac{1}{\sigma^2} X^\top X.$$

Because X is the oriented incidence matrix of the bipartite reviewer–proposal graph,

$$X^\top X = L_G,$$

where L_G denotes its graph Laplacian. The orientation chosen for the edges does not affect this product. For the present simple bipartite graph, the diagonal entries of L_G are the vertex degrees, while an off-diagonal entry is -1 when the corresponding proposal and reviewer are connected by an observed review and zero otherwise.

The hierarchical priors in V1 are

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2), \quad c_j \sim \mathcal{N}(0, \tau_b^2),$$

where the second relation follows from $c_j = -b_j$ and the symmetric prior on b_j . Conditional on τ_θ and τ_b , the prior for β is therefore multivariate normal with zero mean and precision matrix

$$P_0 = \text{diag}(\tau_\theta^{-2} I_{42}, \tau_b^{-2} I_9).$$

Hence

$$p(\beta \mid \tau_\theta, \tau_b) \propto \exp\left(-\frac{1}{2} \beta^\top P_0 \beta\right).$$

Combining likelihood and prior gives

$$p(\beta \mid y, \sigma, \tau_\theta, \tau_b) \propto \exp\left[-\frac{1}{2} \left\{ \beta^\top \left(P_0 + \frac{1}{\sigma^2} X^\top X \right) \beta - 2\beta^\top \frac{X^\top y}{\sigma^2} \right\}\right].$$

This is the kernel of a multivariate normal distribution. Its conditional posterior precision matrix is therefore

$$Q = P_0 + \frac{1}{\sigma^2} X^\top X = P_0 + \frac{1}{\sigma^2} L_G,$$

and its conditional posterior covariance matrix is

$$V = Q^{-1} = \left(P_0 + \frac{1}{\sigma^2} L_G \right)^{-1}.$$

For completeness, the corresponding conditional posterior mean is

$$m = V \frac{X^\top y}{\sigma^2},$$

so that

$$\beta \mid y, \sigma, \tau_\theta, \tau_b \sim \mathcal{N}(m, V).$$

This establishes the statistical role of the assignment graph. Conditional on the variance parameters, the graph Laplacian enters directly into the posterior precision of the proposal and reviewer effects. Adding or removing a reviewer–proposal edge changes L_G and therefore changes the conditional posterior covariance V . The graph is consequently not only a visualization of the review assignments: it describes part of the information structure of the V1 model.

The derivation conditions on σ , τ_θ and τ_b to make this relationship explicit. In the fitted Bayesian model these scale parameters are unknown and are estimated jointly by MCMC, as described in Section 3.6 and Section 9.2.

9.9 Posterior-median sensitivity of the review-design calculation

As a sensitivity analysis, the single-review design calculation was repeated using posterior-median plug-in values for the V1 scale parameters rather than averaging utility over posterior draws. The posterior medians were

$$\tilde{\tau}_\theta = 0.220, \quad \tilde{\tau}_b = 0.157, \quad \tilde{\sigma} = 0.920.$$

For these fixed values, the conditional covariance matrix was constructed once, and the A-, D- and E-optimality reductions were recalculated for all 294 currently unobserved reviewer–proposal assignments.

Criterion	Spearman correlation with posterior-averaged ranking	Top-five overlap
A-optimality	0.873	5 / 5
D-optimality	0.990	4 / 5
E-optimality	0.894	4 / 5

The highest-ranked part of the candidate ordering was therefore stable to the plug-in calculation. Under A-optimality, the same five assignments appeared in

the top five under both approaches. Under D- and E-optimality, four of the five highest-ranked assignments were shared. The lower overall rank correlation for A-optimality indicates that some candidate positions further down the ranking are more sensitive to whether posterior uncertainty in the scale parameters is averaged over or replaced by a single representative value.

The sensitivity calculation does not compare the magnitudes of the posterior-averaged and plug-in utility reductions directly. Because the design criteria are nonlinear functions of the scale parameters, evaluating a criterion at posterior-median parameter values need not reproduce the posterior mean of that criterion. The comparison is therefore used only to assess the robustness of the candidate ordering.

References

- Bailey, R. A., and Peter J. Cameron. 2011. *Using Graphs to Find the Best Block Designs*. <https://arxiv.org/abs/1111.3768>.
- Cao, Jing, S. Lynne Stokes, and Song Zhang. 2010. “A Bayesian Approach to Ranking and Rater Evaluation: An Application to Grant Reviews.” *Journal of Educational and Behavioral Statistics* 35 (2): 194–214. <https://doi.org/10.3102/1076998609353116>.
- Center for Reproducible Science and Research Synthesis. 2026a. *CRS Seed Grants: Four Projects Approved*. University of Zurich. <https://www.crs.uzh.ch/en/news/2026-News/Four-CRS-Seed-Grants-Funded.html>.
- Center for Reproducible Science and Research Synthesis. 2026b. *CRS Seed Grants: Seed Grants for Research Synthesis and Meta-Research*. University of Zurich. <https://www.crs.uzh.ch/en/news/2026-News/CRS-Seed-Grants.html>.
- Cook, Wade D., Boaz Golany, Moshe Kress, Michal Penn, and Tal Raviv. 2005. “Optimal Allocation of Proposals to Reviewers to Facilitate Effective Ranking.” *Management Science* 51 (4): 655–61. <https://doi.org/10.1287/mnsc.1040.0290>.
- Eblen, Matthew K., Robin M. Wagner, Deepshikha RoyChowdhury, Katherine C. Patel, and Katrina Pearson. 2016. “How Criterion Scores Predict the Overall Impact Score and Funding Outcomes for National Institutes of Health Peer-Reviewed Applications.” *PLOS ONE* 11 (6): e0155060. <https://doi.org/10.1371/journal.pone.0155060>.
- Erosheva, Elena A., Sheridan Grant, Mei-Ching Chen, Mark D. Lindner, Richard K. Nakamura, and Carole J. Lee. 2020. “NIH Peer Review: Criterion Scores

- Completely Account for Racial Disparities in Overall Impact Scores.” *Science Advances* 6 (23): eaaz4868. <https://doi.org/10.1126/sciadv.aaz4868>.
- Graves, Nicholas, Adrian G. Barnett, and Philip Clarke. 2011. “Funding Grant Proposals for Scientific Research: Retrospective Analysis of Scores by Members of Grant Review Panel.” *BMJ* 343: d4797. <https://doi.org/10.1136/bmj.d4797>.
- Heyard, Rachel, Manuela Ott, Georgia Salanti, and Matthias Egger. 2022. “Rethinking the Funding Line at the Swiss National Science Foundation: Bayesian Ranking and Lottery.” *Statistics and Public Policy* 9 (1): 110–21. <https://doi.org/10.1080/2330443X.2022.2086190>.
- Heyard, Rachel, David G. Pina, Ivan Buljan, and Ana Marušić. 2025. “Assessing the Potential of a Bayesian Ranking as an Alternative to Consensus Meetings for Decision Making in Research Funding: A Case Study of Marie Skłodowska-Curie Actions.” *PLOS ONE* 20 (3): e0317772. <https://doi.org/10.1371/journal.pone.0317772>.
- Jayasinghe, Upali W., Herbert W. Marsh, and Nigel Bond. 2003. “A Multilevel Cross-Classified Modelling Approach to Peer Review of Grant Proposals: The Effects of Assessor and Researcher Attributes on Assessor Ratings.” *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 166 (3): 279–300. <https://doi.org/10.1111/1467-985X.00278>.
- Johnson, Valen E. 2008. “Statistical Analysis of the National Institutes of Health Peer Review System.” *Proceedings of the National Academy of Sciences of the United States of America* 105 (32): 11076–80. <https://doi.org/10.1073/pnas.0804538105>.
- Kaplan, David, Nicola Lacetera, and Celia Kaplan. 2008. “Sample Size and Precision in NIH Peer Review.” *PLOS ONE* 3 (7): e2761. <https://doi.org/10.1371/journal.pone.0002761>.
- Laird, Nan M., and Thomas A. Louis. 1989. “Empirical Bayes Ranking Methods.” *Journal of Educational Statistics* 14 (1): 29–46. <https://doi.org/10.3102/10769986014001029>.
- Osting, Braxton, Christoph Brune, and Stanley J. Osher. 2014. “Optimal Data Collection for Informative Rankings Expose Well-Connected Graphs.” *Journal of Machine Learning Research* 15: 2981–3012.
- Simpson, Edwin, and Stephen Roberts. 2015. “Bayesian Methods for Intelligent Task Assignment in Crowdsourcing Systems.” In *Scalable Decision Making: Uncertainty, Imperfection, Deliberation*, edited by Tatiana Guy, Miroslav Kárný, and David Wolpert, vol. 538. Studies in Computational Intelligence.

Springer. https://doi.org/10.1007/978-3-319-15144-1_1.

Stelmakh, Ivan, Nihar B. Shah, and Aarti Singh. 2021. "PeerReview4All: Fair and Accurate Reviewer Assignment in Peer Review." *Journal of Machine Learning Research* 22 (163): 1–66.

Tsagris, Michail, Christina Beneki, and Hossein Hassani. 2014. "On the Folded Normal Distribution." *Mathematics* 2 (1): 12–28. <https://doi.org/10.3390/math2010012>.